FINAL DRAWINGS

WITH INSTRUCTION MANUAL FOR PLATE HEAT EXCHANGER

HULL NO. : **H 2511**

CUSTOMER : **HYUNDAI MIPO DOCKYARD CO., LTD.**

DATE : **2014. 09. 22**

Alfa Laval

SPECIFICATION FOR ALFA LAVAL PLATE HEAT EXCHANGER

Customer: HYUNDAI MIPO DOCKYARD CO., LTD.

Hull No. 1 H2511

Application		MAIN L.O COOLER		NO.1 & NO.2 CENTRAL F.W COOLER	
No. of Units		1		2	
Unit Type		MX25-MFMS		T20-PFG	
No. of Plates/Unit		124		193+1	
Heat Transfer Surface/Unit	m2	180.6		170	
Media		L.O	F.W	F.W	S.W
Flowrate	m3/h	220.0	94.0	280.0	320.0
Inlet Temperature	Deg.C.	52.2	36.0	55.1	32.0
Outlet Temperature	Deg.C.	45.0	43.0	36.0	48.9
Pressure Drop	mwg	7.00	0.548	6.61	9.96
Heat Transferred	Kcal/h	653,600		5,280,000	
K-value (Service)	Kcal/m2h.deg.C	397.4		6,221	
K-value (Clean)	Kcal/m2h.deg.C	460.7		7,155	
LMTD	Deg.C	9.1		5.0	
Max. Design Pressure	bar	6	6	6	6
Test Pressure	bar	9	9	9	9
Max. Design Temperature	Deg.C	90	90	90	90
No. of Passages		1	1	1	1
No. of Channels		62MH	61ML	72H+24ML	72H+24MH
Frame Material		MILD STEEL			
Manuf. No.		30115-97558		30115-97559/60	
Plate Material/Thickness	mm	SUS 316 / 0.5 mm		TI / 0.5 mm	
Gasket Material		NBRP			
Fouling margin		15%		15%	
Rule		1 KR (H2511~13, H2517~19, H2547~49), ABS (H2541~46) and LR (H2514~16, H2520~22, H2550~52)			
Colour		1 Munsell No. 7.5 BG 7/2			

Remarks: :

No. 1 / 2

Alfa Laval Korea Ltd.

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Alfa Laval

SPECIFICATION FOR ALFA LAVAL PLATE HEAT EXCHANGER

Customer: HYUNDAI MIPO DOCKYARD CO., LTD.

Hull No. 1 H2511

Application		H.T.F.W COOLER		M/E & G/E MGO COOLER	
No. of Units		1		1	
Unit Type		M10-MFM		M6-FG	
No. of Plates/Unit		34		98	
Heat Transfer Surface/Unit	m2	7.0		14.4	
Media		H.T.F.W	F.W	M.G.O	F.W
Flowrate	m3/h	97.0	94.0	8.0	15.0
Inlet Temperature	Deg.C.	90.0	43.0	52.0	36.0
Outlet Temperature	Deg.C.	78.4	54.7	38.0	39.0
Pressure Drop	mwg	4.93	5.55	0.611	1.37
Heat Transferred	Kcal/h	1,084,000		45,000	
K-value (Service)	Kcal/m2h.deg.C	4,342.0		529.7	
K-value (Clean)	Kcal/m2h.deg.C	7,427.0		609.5	
LMTD	Deg.C	35.4		5.9	
Max. Design Pressure	bar	6	6	11	11
Test Pressure	bar	9	9	16.5	16.5
Max. Design Temperature	Deg.C	90	90	60	60
No. of Passages		1	1	1	1
No. of Channels		17L	16L	48H	49H
Frame Material		MILD STEEL			
Manuf. No.		30115-97561		30115-97562	
Plate Material/Thickness	mm	SUS 304 / 0.4 mm		SUS 316 / 0.5 mm	
Gasket Material		NBRP			
Fouling margin		15%		15%	
Rule		1 KR (H2511~13, H2517~19, H2547~49), ABS (H2541~46) and LR (H2514~16, H2520~22, H2550~52)			
Colour		1 Munsell No. 7.5 BG 7/2			

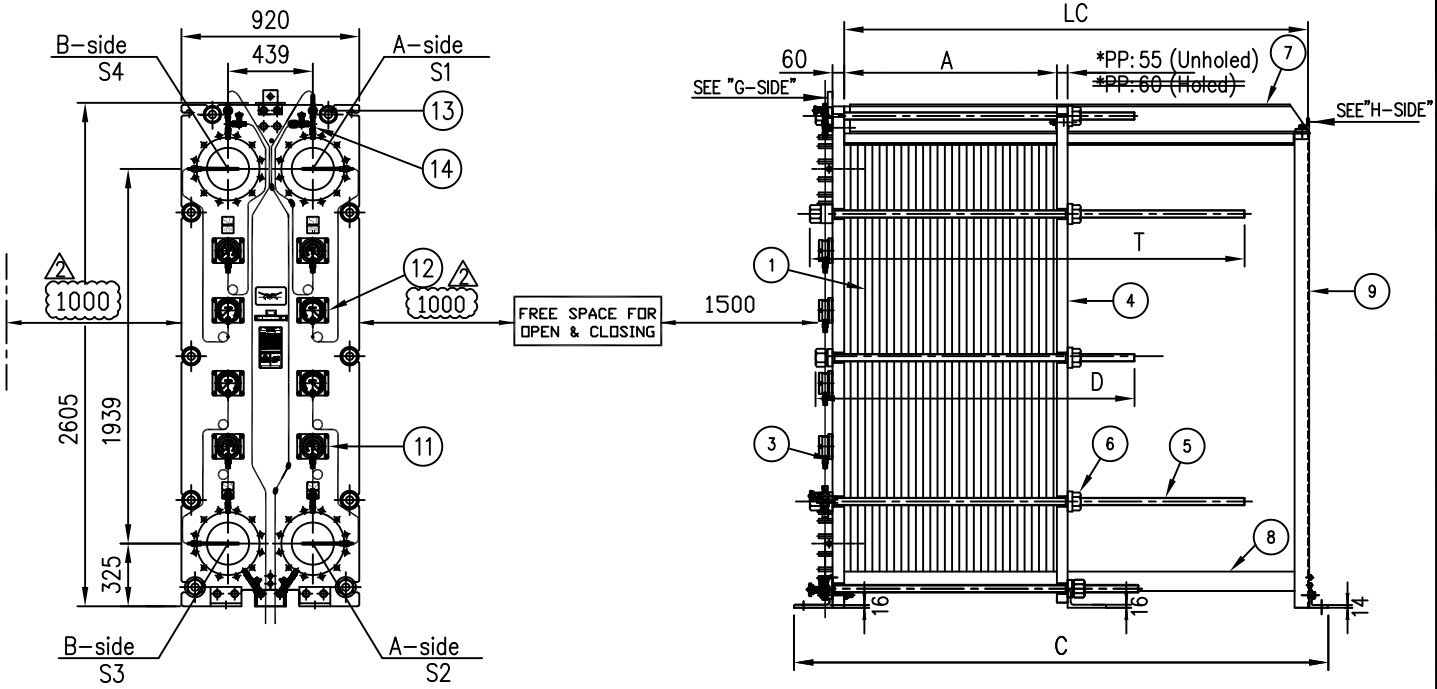
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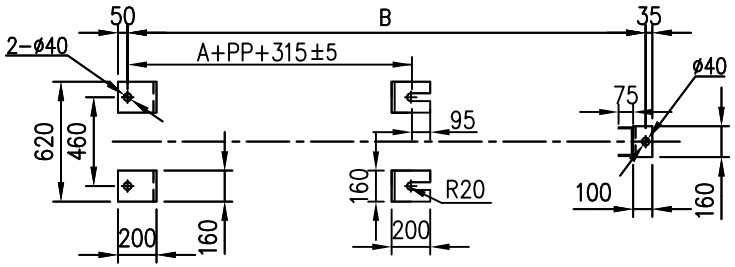
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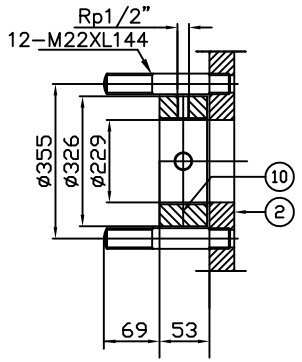
Seoul 100-710, Korea .Telephone: +82 2 3406 0600 Telefax: +82 2 3406 0701





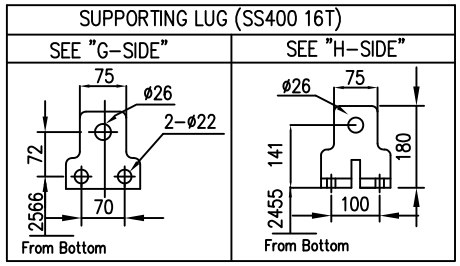
CONNECTIONS

L.O/F.W SIDE(UNLINED)



No. of plate	Dimension(mm)					
	A	LC	B	C	T	D
124	558	1800	2082	2167	1650	1350

SUPPORTING LUG (SS400 16T)



CUSTOMER / REF.	HMD H2511
APPLICATION	MAIN L.O COOLER

CONN. NAME	B-SIDE(F.W)	CONN. NAME	A-SIDE(L.O)
S4	OUT	S1	IN
S3	IN	S2	OUT
CONN. MATERIAL	UNLINED	CONN. MATERIAL	UNLINED
CONNECTION SIZE	JIS10K 250A	CONNECTION SIZE	JIS10K 250A
DESIGN PRESSURE	6 bar	DESIGN PRESSURE	6 bar
TEST PRESSURE	9 bar	TEST PRESSURE	9 bar
LIQUID VOLUME	345.26 L	LIQUID VOLUME	350.84 L

WEIGHT (DRY/WATER FILLED)	3090/3730 KG
PAINT COLOR	⚠1 Munsell No. 7.5 BG 7/2
APP. DRAWING No.	MF2420012-A
DRAWING No.	MF2420012-A
MANUFACTURING NO.	30115-97558

Customer Comment Date

⚠1 2014.03.25 - SJK

⚠2 2014.03.31 - SJK

APPROVAL DATE		2014.01.07	FINAL DATE	2014.09.22 SJK
	Item No.	PART NAME	MATERIAL	Q'ty	Remark
SELECTION PART	①	PLATE/GASKET	SUS316,0.5t/NBRP	124	-
	②	CONNECTION	UNLINED	4	-
	⑩	INSTRUMENT FLANGE	SS400	4	-
	⑪	PRESSURE GAUGE	SUS 316	4	0~10bar
	⑫	THERMOMETER	SUS 316	4	D-type
	⑬	AIR VENT VALVE	C3771	2	-
	⑭	NEEDLE VALAE	C3771	4	-
	⑮	DRAIN VALAE	C3771	2	-
STANDARD PART	③	FRAME PLATE	SS400	1	-
	④	PRESSURE PLATE	SS400	1	-
	⑤	TIGHTENING BOLT	Grade B7	10	M39
	⑥	TIGHTENING NUT	Grade 2H	10	M39
	⑦	CARRYING BAR	SS400+SUS304	1	-
	⑧	GUIDING BAR	SUS 304	1	-
	⑨	SUPPORT COLUMN	SS400	1	-

PLATE HEAT EXCHANGER

MX25-MFMS

THIRD ANGLE

Revion No : ⚠ ALSM-MX25-MFMS

APP'D BY HCJung

CHK'D BY SESong

DRW'N BY CMLee

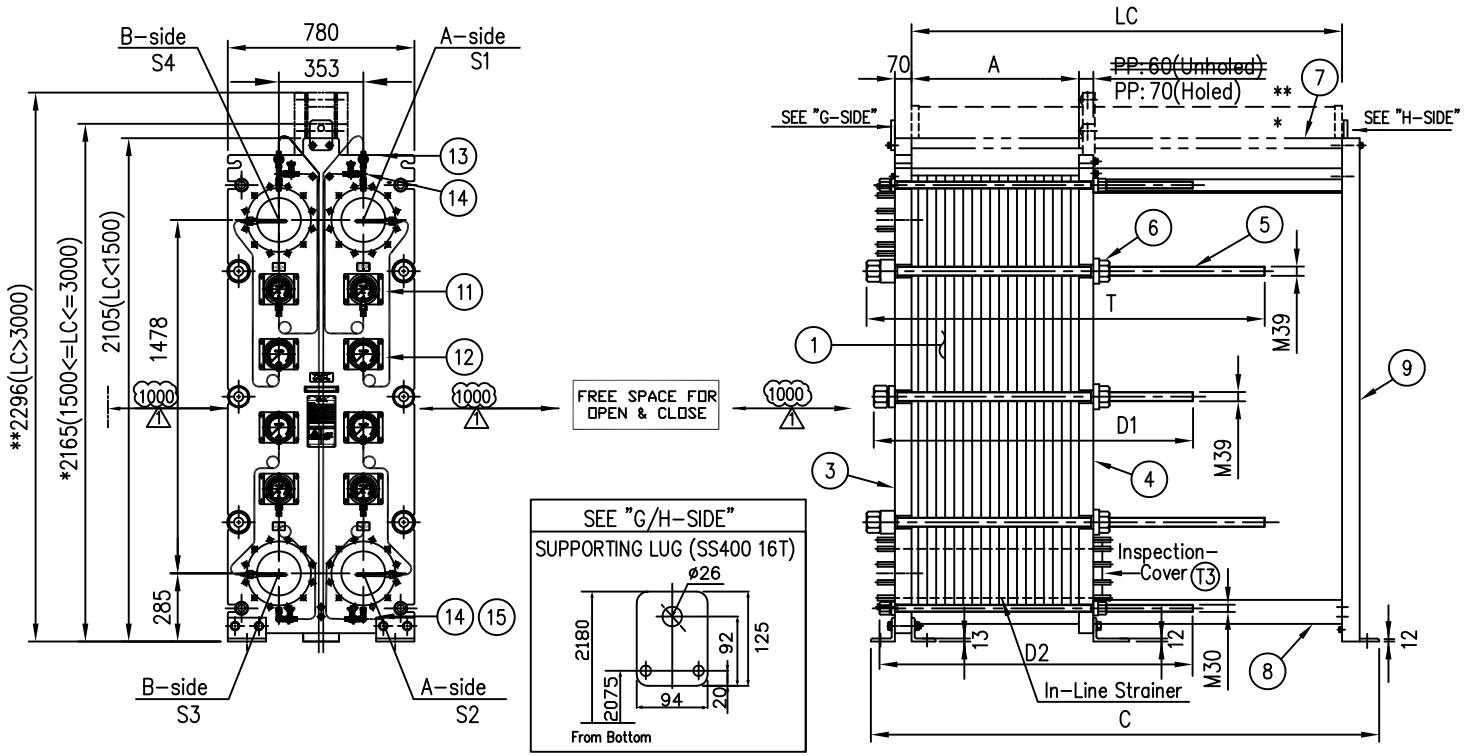
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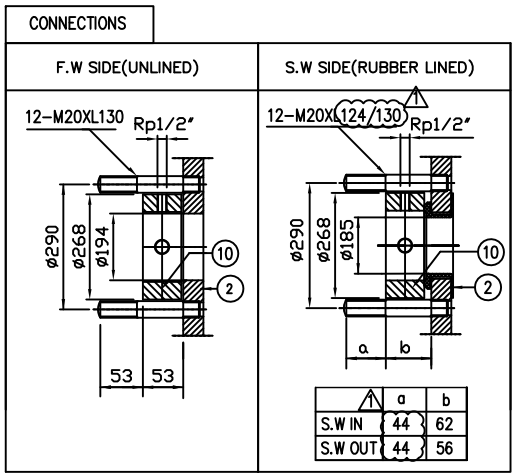
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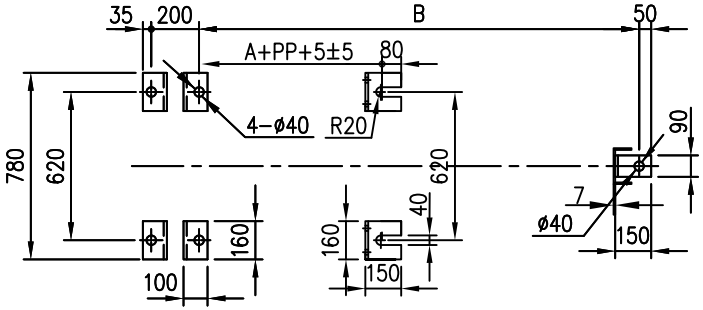
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No. of Plate	Dimension(mm)					
	A	LC	B	C	T	D1/D2
193+1	660	1600	1642	1927	1350	1350

CUSTOMER / REF.	HMD H2511		
APPLICATION	⚠1 NO.1 CENTRAL F.W COOLER		
CONN. NAME	B-SIDE(S.W)	CONN. NAME	A-SIDE(F.W)
S4	OUT	S1	IN
S3	IN	S2	OUT
CONN. MATERIAL	RUBBER	CONN. MATERIAL	UNLINED
CONNECTION SIZE	JIS10K 200A	CONNECTION SIZE	JIS10K 200A
DESIGN PRESSURE	6 bar	DESIGN PRESSURE	6 bar
TEST PRESSURE	9 bar	TEST PRESSURE	9 bar
LIQUID VOLUME	273.80 L	LIQUID VOLUME	273.80 L
WEIGHT (DRY/WATER FILLED)	2200/2750 KG		
PAINT COLOR	⚠2 Munsell No. 7.5 BG 7/2		
APP. DRAWING No.	MF2420012-B1		
DRAWING No.	MF2420012-B1		
MANUFACTURING NO.	30115-97559		

Customer Comment Date

⚠1 2014.03.17 - SJK

⚠2 2014.03.25 - SJK

APPROVAL DATE		2014.01.07	FINAL DATE	2014.09.22 SJK
	Item No.	PART NAME	MATERIAL	Q'ty	Remark
SELECTION PART	①	PLATE/GASKET	TI, 0.5t/NBRP	194	-
	②	CONNECTION	UNLINED/RUBBER	2/2	-
	⑩	INSTRUMENT FLANGE	SS400/C4640	2/2	-
	⑪	PRESSURE GAUGE	SUS 316	4	0~10bar
	⑫	THERMOMETER	SUS 316	4	D-type
	⑬	AIR VENT VALVE	C3771/C4640	1/1	-
	⑭	NEEDLE VALAE	C3771/C4640	2/2	-
	⑮	DRAIN VALAE	C3771/C4640	1/1	-
STANDARD PART	③	FRAME PLATE	SS400	1	-
	④	PRESSURE PLATE	SS400	1	-
	⑤	TIGHTENING BOLT	Grade B7	6/4	M39/M30
	⑥	TIGHTENING NUT	Grade 2H	6/4	M39/M30
	⑦	CARRYING BAR	AL(SUS LINING)/ **SS400+SUS304	1	-
	⑧	GUIDING BAR	SUS 304	1	-
	⑨	SUPPORT COLUMN	SS400	1	-

PLATE HEAT EXCHANGER

T20-PFG

THIRD ANGLE

Revion No : ⚠ ALSM-T20-PFG

APP'D BY HCJung

CHK'D BY SESong

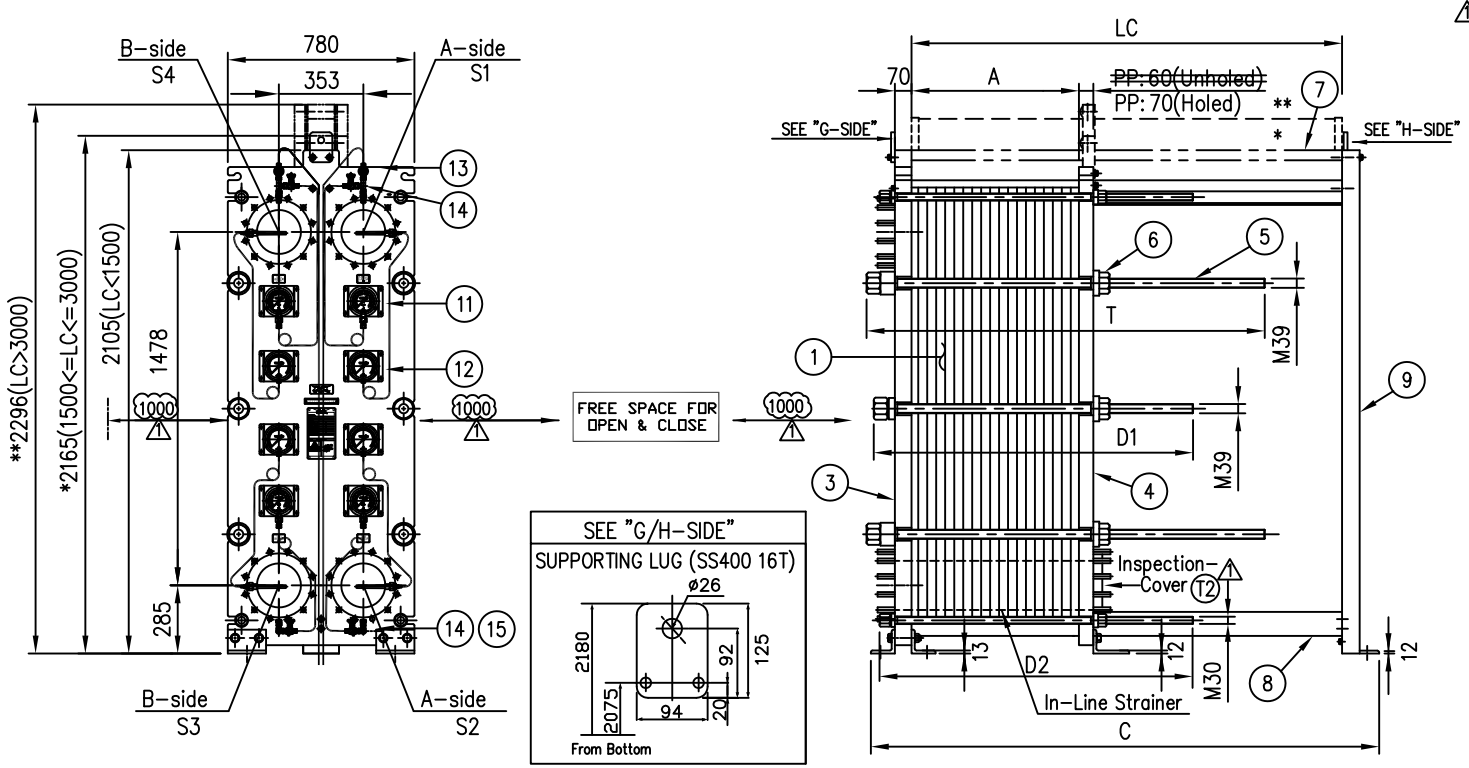
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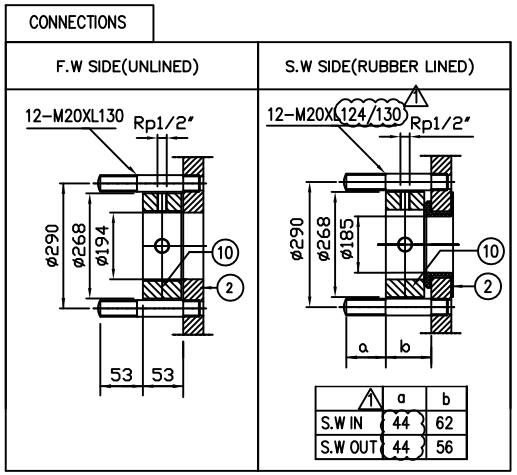
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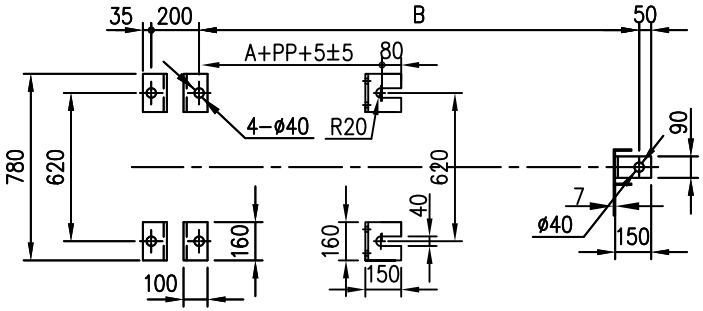
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No. of Plate	Dimension(mm)					
	A	LC	B	C	T	D1/D2
193+1	660	1600	1642	1927	1350	1350

CUSTOMER / REF.	HMD H2511
APPLICATION	NO.2 CENTRAL F.W COOLER

CONN. NAME	B-SIDE(F.W)	CONN. NAME	A-SIDE(S.W)
S4	IN	S1	OUT
S3	OUT	S2	IN
CONN. MATERIAL	UNLINED	CONN. MATERIAL	RUBBER
CONNECTION SIZE	JIS10K 200A	CONNECTION SIZE	JIS10K 200A
DESIGN PRESSURE	6 bar	DESIGN PRESSURE	6 bar
TEST PRESSURE	9 bar	TEST PRESSURE	9 bar
LIQUID VOLUME	273.80 L	LIQUID VOLUME	273.80 L

WEIGHT (DRY/WATER FILLED)	2200/2750 KG
PAINT COLOR	Munsell No. 7.5 BG 7/2
APP. DRAWING No.	⚠2 MF2420012-B2
DRAWING No.	MF2420012-B2
MANUFACTURING NO.	30115-97560

Customer Comment Date

⚠1 2014.03.17 - SJK

⚠2 2014.03.25 - SJK

APPROVAL DATE		2014.01.07	FINAL DATE	2014.09.22 SJK
	Item No.	PART NAME	MATERIAL	Q'ty	Remark
SELECTION PART	①	PLATE/GASKET	Ti, 0.5t/NBRP	194	-
	②	CONNECTION	UNLINED/RUBBER	2/2	-
	⑩	INSTRUMENT FLANGE	SS400/C4640	2/2	-
	⑪	PRESSURE GAUGE	SUS 316	4	0~10bar
	⑫	THERMOMETER	SUS 316	4	D-type
	⑬	AIR VENT VALVE	C3771/C4640	1/1	-
	⑭	NEEDLE VALAE	C3771/C4640	2/2	-
	⑮	DRAIN VALAE	C3771/C4640	1/1	-
STANDARD PART	③	FRAME PLATE	SS400	1	-
	④	PRESSURE PLATE	SS400	1	-
	⑤	TIGHTENING BOLT	Grade B7	6/4	M39/M30
	⑥	TIGHTENING NUT	Grade 2H	6/4	M39/M30
	⑦	CARRYING BAR	AL(SUS LINING)/ **SS400+SUS304	1	-
	⑧	GUIDING BAR	SUS 304	1	-
	⑨	SUPPORT COLUMN	SS400	1	-

PLATE HEAT EXCHANGER

T20-PFG

THIRD ANGLE

Revion No : ⚠1 ALSM-T20-PFG

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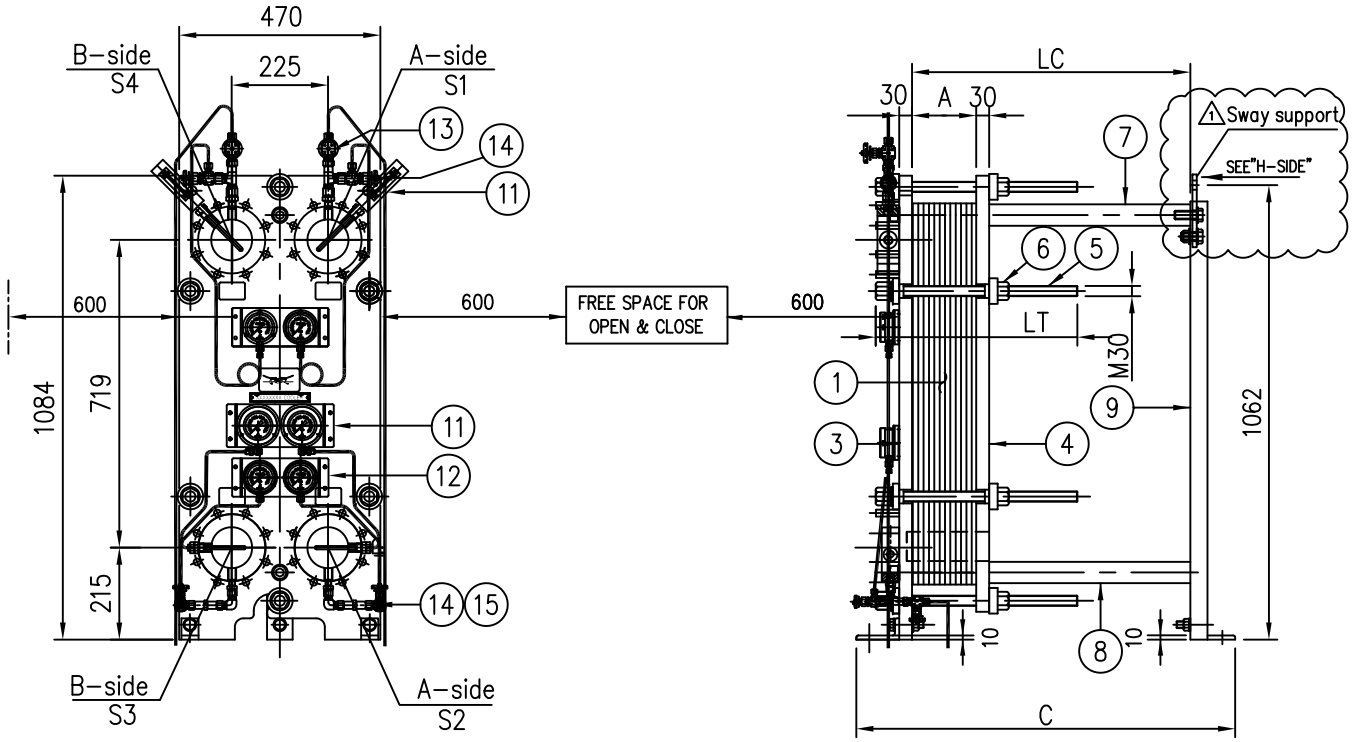
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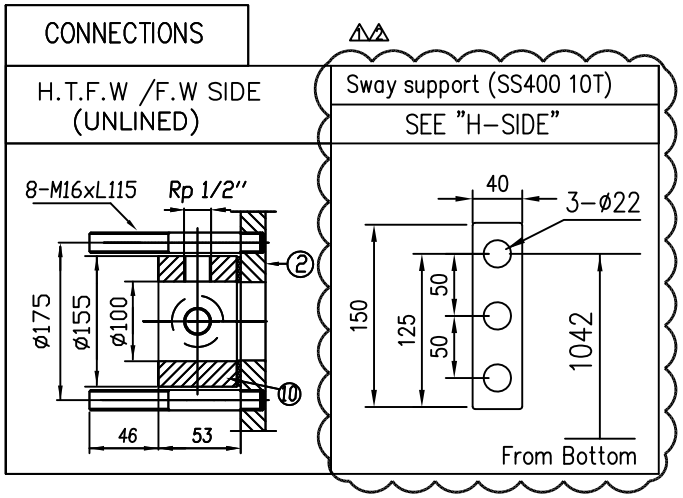
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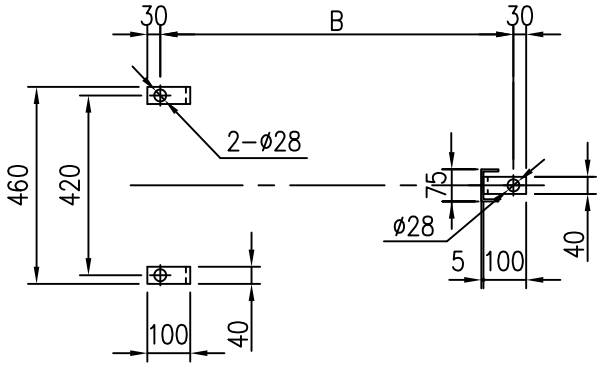
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No. of plate	Dimension(mm)				
	A	LC	B	C	LT
34	152	650	825	885	450

CUSTOMER / REF.	HMD H2511		
APPLICATION	H.T.F.W COOLER		
CONN. NAME	B-SIDE(H.T.F.W)	CONN. NAME	A-SIDE(F.W)
S4	IN	S1	OUT
S3	OUT	S2	IN
CONN. MATERIAL	UNLINED	CONN. MATERIAL	UNLINED
CONNECTION SIZE	JIS10K 100A	CONNECTION SIZE	JIS10K 100A
DESIGN PRESSURE	6 bar	DESIGN PRESSURE	6 bar
TEST PRESSURE	9 bar	TEST PRESSURE	9 bar
LIQUID VOLUME	17.46 L	LIQUID VOLUME	16.46 L
WEIGHT (DRY/WATER FILLED)		308/340	KG
PAINT COLOR		Munsell No. 7.5 BG 7/2	
APP. DRAWING No.		MF2420012-C	
DRAWING No.		MF2420012-C	
MANUFACTURING NO.		30115-97561	

Customer Comment Date

2014.03.25 - SJK

2014.08.19 - SJK

APPROVAL DATE		2014.01.07	FINAL DATE	2014.09.22 SJK
	Item No.	PART NAME	MATERIAL	Q'ty	Remark
SELECTION PART	①	PLATE/GASKET	SUS304,0.4t/NBRP	34	-
	②	CONNECTION	UNLINED	4	-
	⑩	INSTRUMENT FLANGE	SS400	4	-
	⑪	PRESSURE GAUGE	SUS 316	4	0~10bar
	⑫	THERMOMETER	SUS 316	2/2	I-type D-type
	⑬	Air vent Valve	C3771	2	-
	⑭	NEEDLE VALAE	C3771	4	-
	⑮	DRAIN VALVE	C3771	2	
STANDARD PART	③	FRAME PLATE	SS400	1	-
	④	PRESSURE PLATE	SS400	1	-
	⑤	TIGHTENING BOLT	Grade B7	6	M30
	⑥	TIGHTENING NUT	Grade 2H	6	M30
	⑦	CARRYING BAR	SGD 400	1	-
	⑧	GUIDING BAR	SGD 400	1	-
	⑨	SUPPORT COLUMN	SS400	1	-

PLATE HEAT EXCHANGER

M10-MFM

THIRD ANGLE

Revion No : ALSM-M10-MFM

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CHK'D BY SESong

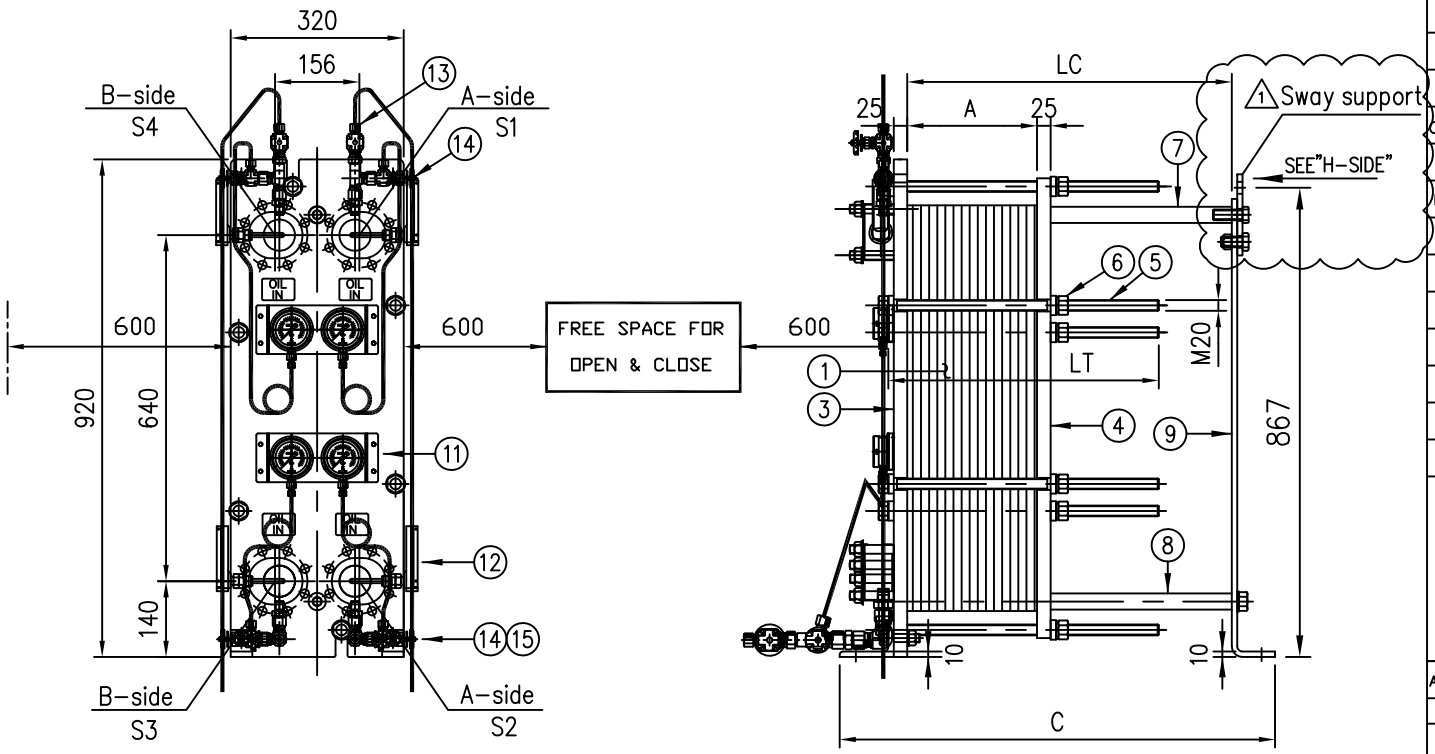
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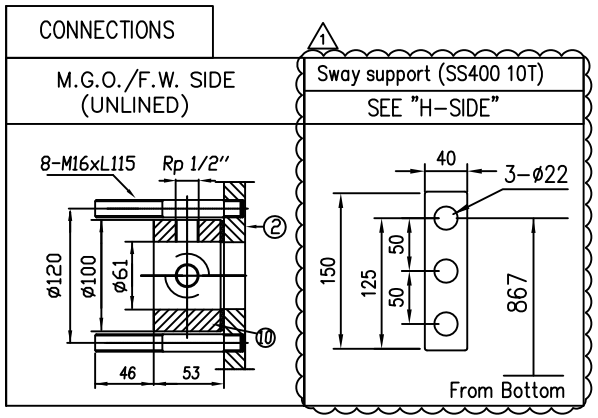
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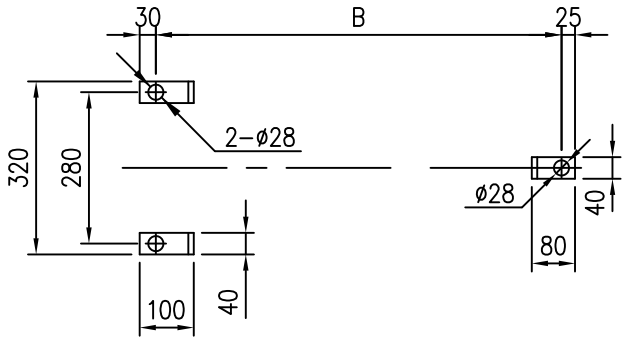
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No. of plate	Dimension (mm)				
	A	LC	B	C	LT
98	245	750	900	955	500

CUSTOMER / REF.	HMD H2511		
APPLICATION	M/E & G/E MGO COOLER		
CONN. NAME	B-SIDE(F.W.)	CONN. NAME	A-SIDE(M.G.O.)
S4	OUT	S1	IN
S3	IN	S2	OUT
CONN. MATERIAL	UNLINED	CONN. MATERIAL	UNLINED
CONN. SIZE	JIS 16K 50A	CONN. SIZE	JIS 16K 50A
DESIGN PRESSURE	11 bar	DESIGN PRESSURE	11 bar
TEST PRESSURE	16.5 bar	TEST PRESSURE	16.5 bar
LIQUID VOLUME	14.86 L	LIQUID VOLUME	14.56 L
WEIGHT (DRY/WATER FILLED)	212/239 KG		
PAINT COLOR	⚠1 Munsell No. 7.5 BG 7/2		
APP. DRAWING No.	MF2420012-D		
DRAWING No.	MF2420012-D		
MANUFACTURING NO.	30115-97562		

Customer Comment Date
⚠1 2014.03.25 - SJK

APPROVAL DATE	2014.01.07	FINAL DATE	2014.09.22 SJK	
	Item No.	PART NAME	MATERIAL	Q'ty	Remark
SELECTION PART	①	PLATE/GASKET	SUS316,0.5t/NBRP	98	-
	②	CONNECTION	UNLINED	4	-
	⑩	INSTRUMENT FLANGE	SS400	4	-
	⑪	PRESSURE GAUGE	SUS 316	4	0~20bar
	⑫	THERMOMETER	SUS 316	4	L-type
	⑬	AIR VENT VALVE	C3771	2	-
	⑭	NEEDLE VALAE	C3771	4	-
	⑮	DRAIN VALVE	C3771	2	-
STANDARD PART	③	FRAME PLATE	SS400	1	-
	④	PRESSURE PLATE	SS400	1	-
	⑤	TIGHTENING BOLT	Grade B7	6	M20
	⑥	TIGHTENING NUT	Grade 2H	6	M20
	⑦	CARRYING BAR	SGD 400	1	-
	⑧	GUIDING BAR	SGD 400	1	-
	⑨	SUPPORT COLUMN	SS400	1	-

PLATE HEAT EXCHANGER
M6-FG

THIRD ANGLE

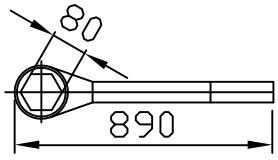
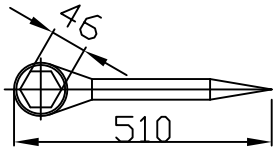
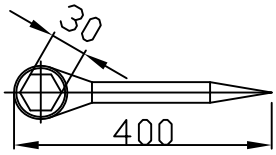
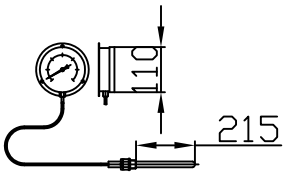
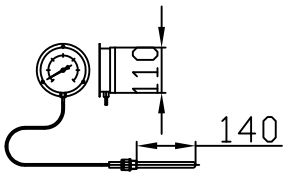
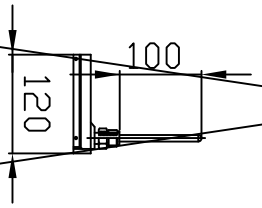
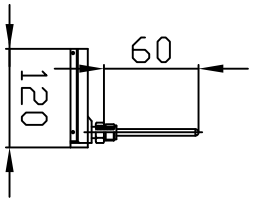
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ALSM-M6-FG

APP'D BY HCJung
CHK'D BY SESong
DRW'N BY CMLee
DATE 100118

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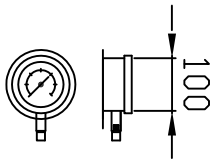
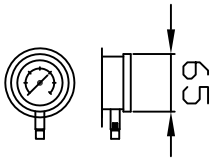
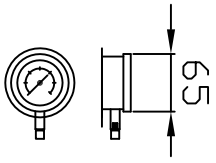
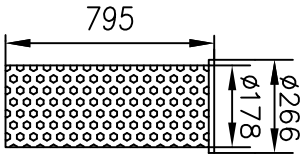
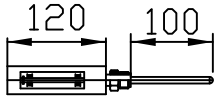
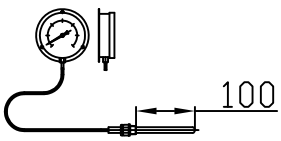
TOOL/SPARE/ACCESSORY						PAGE		1/2
ALFA LAVAL PLATE HEAT EXCHANGE						SHIP NO		HMD H2511's
TOOL/SPARE: MX25-MFMS x 1, T20-PFG x 2, M10-MFM x 1, M6-FG						BOX NO		2511
NO	NAME	SKETCH	MATERIAL	SUPPLY/SHIP		DRAWING		REMARK
				WORKING	SPARE	Q'ty	PART NO.	
1	SPANNER		TOOL STEEL	-	3		32220-3195-1	FOR MX25-MFMS T20-PFG
2	SPANNER		TOOL STEEL	-	3		32800419-01	FOR T20-PFG M10-MFM
3	SPANNER		TOOL STEEL	-	1		32800-5790-2	FOR M6-FG
4	THERMOMETER		SUS 316	4	-		32860098-02	DIAL TYPE 0~100℃ LEAD LENGTH: 1.5M FOR MX25-MFMS
5	THERMOMETER		SUS 316	8	-		32860098-01	DIAL TYPE 0~100℃ LEAD LENGTH: 1.5M FOR T20-PFG
6	THERMOMETER		SUS 316	4	-		32860024-01	L-TYPE 0~100℃ FOR M10-MFM
7	THERMOMETER		SUS 316	4	-		32860024-02	L-TYPE 0~100℃ FOR M6-FG

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TOOL/SPARE/ACCESSORY								PAGE	2/2
ALFA LAVAL PLATE HEAT EXCHANGE								SHIP NO	HMD H2511's
TOOL/SPARE: MX25-MFMS x 1, T20-PFG x 2, M10-MFM x 1, M6-FG								BOX NO	2511

NO	NAME	SKETCH	MATERIAL	SUPPLY/SHIP		DRAWING		REMARK
				WORKING	SPARE	Q'ty	PART NO.	
8	PRESSURE GAUGE		SUS 316	12	–		32860021-05	RANGE: 0~10 bar OIL FILLED TYPE fitted on body MX25-MFMS T20-PFG
9	PRESSURE GAUGE		SUS 316	4	–		32860022-05	RANGE: 0~10 bar OIL FILLED TYPE fitted on body M10-MFM
10	PRESSURE GAUGE		SUS 316	4	–		32860022-05	RANGE: 0~20 bar OIL FILLED TYPE fitted on body M6-FG
11	IN-LINE STRAINER (Filter + Guide Ring)		254 SMO	2	–		32860601-06	FOR T20-PFG (CENT.)
12	THERMOMETER		SUS 316	2	–		32860023-01	I-TYPE FOR M10-MFM
13	THERMOMETER		SUS 316	2 (2)	–		328602326-1	DIAL TYPE 0~100℃ LEAD LENGTH: 0.75M FOR M10-MFM

(1)

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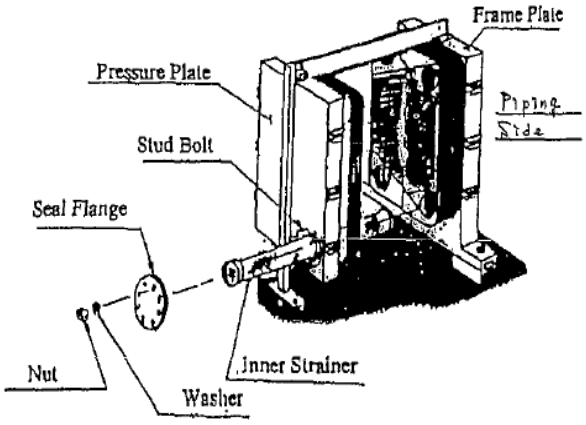
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IN-LINE FILTER FOR ALFA LAVAL PLATE HEAT EXCHANGER

- The In-line filter is that a tube of stainless steel providing which to be installed inside Sea Water passage of Plate Heat Exchanger to prolong the cleaning interval of PHE.
- The In-line filter can be cleaned by dismantling of Sea Flange installed on Pressure Plate without disassembling of PHE.
- The In-line filter is not included on our standard scope of supply. (Option)
- Assembling / disassembling and cleaning of the In-Line filter to be carried out as follows;
  1. Stop Sea Water pump and L.O. / F.W. pumps.
     Liquid drain in the PHE to be discharged thoroughly by the drain valves and the air vent valves.
  2. The Sea Flange on the Pressure plate for Sea water passage to be dismantled. And then the In-line filter can be pulled out easily by hand.
  3. Clean up the In-line filter by nylon brush in order not to damage the In-Line filter body.
  4. Put the In-line filter into Sea Water passage of PHE after confirm that it is in order.
  5. Assemble the Sea Flange onto the Pressure Plate, and drain valves on the Instrument Flange to be closed.
  6. Air vent valves to be closed after air-purge.

Please refer to the Instruction Manual for disassembling and assembling of PHE.



Note) : If it is possible to back-flush in Sea Water side of the PHE, the In-line filter should be removed from the PHE before back-flushing.
The In-line filter body is crushed by Hydraulic (water) pressure due to that the Sea water flows from outside to inside of the In-line filter body at back-flushing.

Plate Listing

Date : 02/05/2014

Customer : HMD
Model : MX25-MFMS
Project : H2511 **Item : MAIN L.O.CLR**
Manuf. No. : 30115-97558

	Hot side	Cold side
Plate Fam. / Plate Grp / Frame / Plate material / thickness	MX25 / M / FMS / ALLOY 316 / 0.50 mm	
Grouping	1*62MH	1*61ML
Sealing material	NBRP CLIP-AD	NBRP CLIP-AD
Nozzle orientation	S1 -> S2	S4 <- S3

The plates are assembled with the gasket side facing the frame plate.

Plate No		Plate Code No	A/B	1	2	3	4	Plate Pattern
		FramePlate		O	O	O	O	
1		3950399883	A	O	O	O	O	MX25 M2
2		3950400803	B	U --->---	U	O	O	MX25 M2
3		3950400703	A	O	O	U ==>==	U	MX25 M1
4	6...120	3950400803	B					MX25 M2
5	7...121	3950400703	A					MX25 M1
122		3950400803	B	U --->---	U	O	O	MX25 M2
123		3950400703	A	O	O	U ==>==	U	MX25 M1
124		3950399876	B	-- --->---	--			MX25 M2
		PressurePlate						

Article No.	Quant.
3950399883	1
3950400803	61
3950400703	61
3950399876	1

Total	124

Plate Listing

Date : 02/05/2014

Customer : HMD
Model : T20-PFG
Project : H2511 **Item : NO.1 CENTRAL F.W.CLR**
Manuf. No. : 30115-97559

	Hot side	Cold side
Plate Fam. / Plate Grp / Frame / Plate material / thickness	T20 / P / FG / TI / 0.50 mm	
Grouping	1* (72H+24ML)	1* (72H+24MH)
Sealing material	NBRP CLIP-ON	NBRP CLIP-ON
Nozzle orientation	S1 -> S2	S4 <- S3

The plates are assembled with the gasket side facing the frame plate.

Plate No		Plate Code No	A/B	1	2	3	4	Plate Pattern
		FramePlate		O	O	O	O	
1		3951172183	A	O	O	O	O	T20 P2
2		3950724303	B	U --->---	U	O	O	T20 P2
3		3950724303	A	O	O	U ==>==	U	T20 P2
4	6...142	3950724303	B					T20 P2
5	7...143	3950724303	A					T20 P2
144		3950724303	B	U --->---	U	O	O	T20 P2
145		3950724303	A	O	O	U ==>==	U	T20 P2
146		3950724203	B	U --->---	U	O	O	T20 P1
147	149...189	3950724303	A					T20 P2
148	150...190	3950724203	B					T20 P1
191		3950724303	A	O	O	U ==>==	U	T20 P2
192		3950724203	B	U --->---	U	O	O	T20 P1
193		3951172113	A			U ==>==	=	T20 P2
194		3951172147	B			O		T20 P2
		PressurePlate				O		

Article No.	Quant.
3951172183	1
3950724303	167
3950724203	24
3951172113	1
3951172147	1

Total	194

Plate Listing

Date : 02/05/2014

Customer : HMD
Model : T20-PFG
Project : H2511 **Item : NO.2 CENTRAL F.W.CLR**
Manuf. No. : 30115-97560

	Hot side	Cold side
Plate Fam. / Plate Grp / Frame / Plate material / thickness	T20 / P / FG / TI / 0.50 mm	
Grouping	1* (72H+24ML)	1* (72H+24MH)
Sealing material	NBRP CLIP-ON	NBRP CLIP-ON
Nozzle orientation	S4 -> S3	S1 <- S2

The plates are assembled with the gasket side facing the frame plate.

Plate No		Plate Code No	A/B	1	2	3	4	Plate Pattern
		FramePlate		O	O	O	O	
1		3951172183	B	O	O	O	O	T20 P2
2		3950724303	A	O	O	U ---<---	U	T20 P2
3		3950724303	B	U ==<==	U	O	O	T20 P2
4	6...142	3950724303	A					T20 P2
5	7...143	3950724303	B					T20 P2
144		3950724303	A	O	O	U ---<---	U	T20 P2
145		3950724303	B	U ==<==	U	O	O	T20 P2
146		3950724203	A	O	O	U ---<---	U	T20 P1
147	149...189	3950724303	B					T20 P2
148	150...190	3950724203	A					T20 P1
191		3950724303	B	U ==<==	U	O	O	T20 P2
192		3950724203	A	O	O	U ---<---	U	T20 P1
193		3951172108	B	= ==<==	U			T20 P2
194		3951172152	A		O			T20 P2
		PressurePlate			O			

Article No.	Quant.
3951172183	1
3950724303	167
3950724203	24
3951172108	1
3951172152	1

Total	194

Plate Listing

Date : 02/05/2014

Customer : HMD
Model : M10-MFM
Project : H2511 **Item : HTFW CLR**
Manuf. No. : 30115-97561

	Hot side	Cold side
Plate Fam. / Plate Grp / Frame / Plate material / thickness	M10 / M / FM / ALLOY 304 / 0.40 mm	
Grouping	1*17L	1*16L
Sealing material	NBRP CLIP-ON	NBRP CLIP-ON
Nozzle orientation	S4 -> S3	S1 <- S2

The plates are assembled with the gasket side facing the frame plate.

Plate No		Plate Code No	A/B	1	2	3	4	Plate Pattern
		FramePlate		O	O	O	O	
1		367317 0483	B	O	O	O	O	M10 M2
2		367220 2403	A	O	O	U ---<---	U	M10 M1
3		367220 2403	B	U ==<==	U	O	O	M10 M1
4	6...30	367220 2403	A					M10 M1
5	7...31	367220 2403	B					M10 M1
32		367220 2403	A	O	O	U ---<---	U	M10 M1
33		367220 2403	B	U ==<==	U	O	O	M10 M1
34		367317 0476	A			-- ---<---	--	M10 M2
		PressurePlate						

Article No.	Quant.
367317 0483	1
367220 2403	32
367317 0476	1

Total	34

Plate Listing

Date : 02/05/2014

Customer : HMD
Model : M6-FG
Project : H2511 **Item : M/E & G/E MGO CLR**
Manuf. No. : 30115-97562

	Hot side	**Cold side**
Plate Fam. / Plate Grp / Frame / Plate material / thickness	M6 / / FG / ALLOY 316 / 0.50 mm	
Grouping	1*48H	1*49H
Sealing material	NBRP CLIP-ON	NBRP CLIP-ON
Nozzle orientation	S1 -> S2	S4 <- S3

The plates are assembled with the gasket side facing the frame plate.

Plate No		Plate Code No	A/B	1	2	3	4	Plate Pattern
		FramePlate		O	O	O	O	
1		367117 4483	B	O	O	O	O	M62
2		367117 4403	A	O	O	U ==>==	U	M62
3		367117 4403	B	U --->---	U	O	O	M62
4	6...94	367117 4403	A					M62
5	7...95	367117 4403	B					M62
96		367117 4403	A	O	O	U ==>==	U	M62
97		367117 4403	B	U --->---	U	O	O	M62
98		367117 4476	A			= ==>==	=	M62
		PressurePlate						

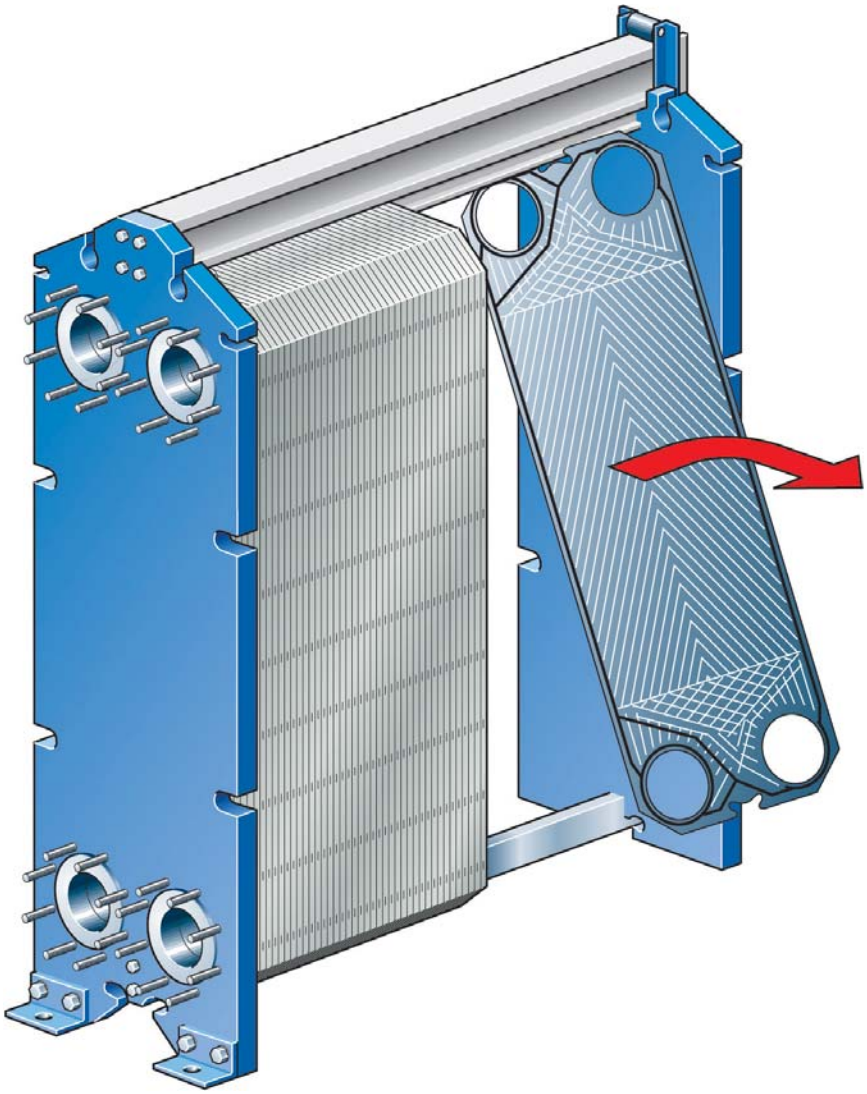
Article No.	Quant.
367117 4483	1
367117 4403	96
367117 4476	1

Total	98

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Instruction Manual
Plate Heat Exchangers

M15 TL10 TL15 T20 TS20 MX25 M30 MA30



Part number 34500621-01
1002

Table of Contents

EN

Description 5
- Main components 5
- Function 6
- Identification of plate side 6
- Semi-welded PHE 6

Installation 7
- Before installation 7
- Requirements 7
- Lifting 8
- Raising 9

Operation 10
- Start-up 10
- Unit in operation 11
- Shut-down 11

Maintenance 12
- Cleaning-In-Place (CIP) 12
- Manual cleaning 13
- Closing 16
- Closing - TL15-B 18
- Pressure test after maintenance 19
- Regasketing 19

How to contact Alfa Laval

Contact details for all countries are continually updated on our website.

Please visit **www.alfalaval.com** and contact your local Alfa Laval Representative.

EN

Preface

This manual provides information needed to install, operate and carry out the maintenance of the Plate Heat Exchanger (PHE).

Safety considerations

The Plate Heat Exchanger shall be used and maintained in accordance with Alfa Laval's instructions in this manual. Faulty handling of the Plate Heat Exchanger may result in serious consequences with injuries to persons and/or property damage. Alfa Laval will not accept responsibility for any damage or injury that has resulted from not following the instructions in this manual.

The Plate Heat Exchanger shall be used in accordance with the specified configuration of material, media types, temperatures and pressure for the specific Plate Heat Exchanger.

The following models are covered in this manual:

- M15
- TL10
- TL15
- T20
- TS20
- MX25
- M30
- MA30
- TL35

Definitions of expressions

Warning!
Type of hazard
WARNING indicates a potentially hazardous situation that, if not avoided, could result in death or serious injury.

Caution!
Type of hazard
CAUTION indicates a potentially hazardous situation that, if not avoided, may result in minor or moderate injury.

Note!
NOTE indicates a potentially hazardous situation that, if not avoided, may result in property damage.

PHE drawings

The PHE drawings mentioned in the manual are the drawings included in the delivery of the Plate Heat Exchanger.

Warranty conditions

The warranty conditions are usually included in the signed sales contract prior to the order of the delivered PHE. Alternatively, the warranty conditions are included in the sales offer documentation or with a reference to the document specifying the valid conditions. If faults occur during the specified warranty period, always consult your local Alfa Laval Representative for advice.

Report the date when the Plate Heat Exchanger was put into operation to the local Alfa Laval Representative.

Advice

Always consult your local Alfa Laval Representative for advice on:

- New plate pack dimensions if you intend to change the number of plates
- Selection of gasket material if operating temperatures and pressures are permanently changed, or if another medium is to be processed in the PHE.

EN

Storage of the PHE

Alfa Laval delivers the PHE ready to be put into service upon arrival if nothing else has been agreed. However, keep the PHE in the packing box until installation.

Regarding storage for longer periods of time, one month or more, certain precautions should be made to avoid unnecessary damage to the PHE.

> ***Note!***
> Alfa Laval and its representatives reserve the right to inspect the storage space and/or equipment whenever necessary until the date of expiry of the warranty period stipulated in the contract. Notification has to be given 10 days prior to the date of inspection.

If there is any uncertainty about the storage of the PHE, consult an Alfa Laval Representative.

Storage in packing box

If the nature of storage after the delivery of the PHE is known in advance, inform Alfa Laval when ordering the PHE to ensure that it will be properly prepared for storage before packing.

Indoor storage

- Store inside a room with the temperature between 15 and 20°C (60 - 70°F) and humidity around 70%. For outdoor storage read “Outdoor storage” on this page.
- To prevent damage to the gaskets, there should not be any ozone-producing equipment in the room such as electric motors or welding equipment.
- To prevent damage to the gaskets, do not store organic solvents or acids in the room and avoid direct sunlight, intensive heat radiation or ultraviolet radiation.
- The tightening bolts should be well covered with light grease coating.

Outdoor storage

If the PHE has to be stored outdoors, all the precautions mentioned in the section “Indoor storage” on this page should be taken. Also, protection against climate is very important.

The stored PHE shall be visually checked every third month. The check includes:

- Greasing of the tightening bolts
- Metal port covers
- Protection of the plate pack and gaskets

Taken out of service

If, for any reason, the PHE is shut down and taken out of service for a long period of time, follow the same advice as in the previous section “Indoor storage” on this page. Although before storage following actions has to be done.

- Check the measurement of the plate pack (measure between frame and pressure plate, A dimension).
- Drain both media sides of the PHE.
- Depending on the media, the PHE should be rinsed and then dried.
- The connection should be covered if the piping system is not connected. Use a plastic or plywood cover for the connection.
- Cover the plate pack with non-transparent plastic film.

Installation after long-term storage

In cases when the PHE has been taken out of service for an extensive period of time, i.e. longer than one year, the risk of leakage when starting up increases. To avoid this problem it is recommended to let the gasket rubber rest and regain most of its elasticity.

1. If the PHE is not in position, follow the instructions “Installation” on page 7.
2. Note the measurement between frame and pressure plate (A dimension).
3. Remove feet attached to the pressure plate.
4. Loosen the tightening bolts. Follow the instructions “Opening” on page 13. Open the PHE until the measure is 1.25A.
5. Leave the PHE between 24-48 hours, the longer the better, for gaskets to relax.
6. Re-tighten according to the instructions “Closing” on page 16 or “Closing - TL15-B” on page 18.
7. Alfa Laval recommends a hydraulic test to be carried out. The media, usually water, should be entered at intervals to avoid sudden shocks to the PHE. It is recommended to test up to the Design Pressure, refer to PHE drawing.

EN

Environmental compliance

Alfa Laval endeavours to perform its own operations as cleanly and efficiently as possible, and to take environmental aspects into consideration when developing, designing, manufacturing, servicing and marketing its products.

Unpacking

Packing material consists of wood, plastics, cardboard boxes and, in some cases, metal straps.

- Wood and cardboard boxes can be reused, recycled or used for energy recovery.
- Plastics should be recycled or burnt at a licensed waste incineration plant.
- Metal straps should be sent for material recycling.

Maintenance

- During maintenance, oil and wear parts in the machine are replaced.
- All metal parts should be sent for material recycling.
- Worn out or defective electronic parts should be sent to a licensed handler for material recycling.
- Oil and all non-metal wear parts must be taken care of in agreement with local regulations.

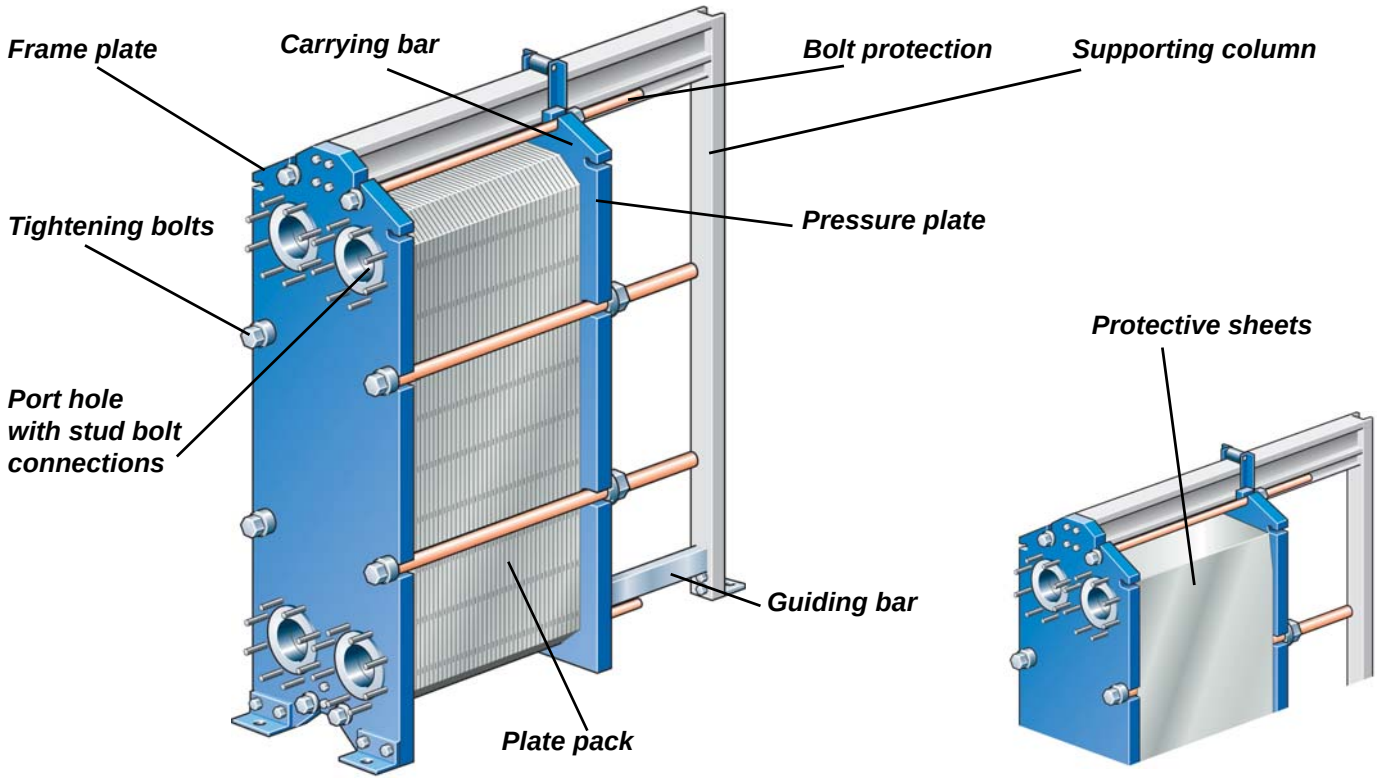
Scrapping

At end of use, the equipment shall be recycled according to relevant, local regulations. Besides the equipment itself, any hazardous residues from the process liquid must be considered and dealt with in a proper manner. When in doubt, or in absence of local regulations, please contact the local Alfa Laval sales company.

EN

Description

Main components



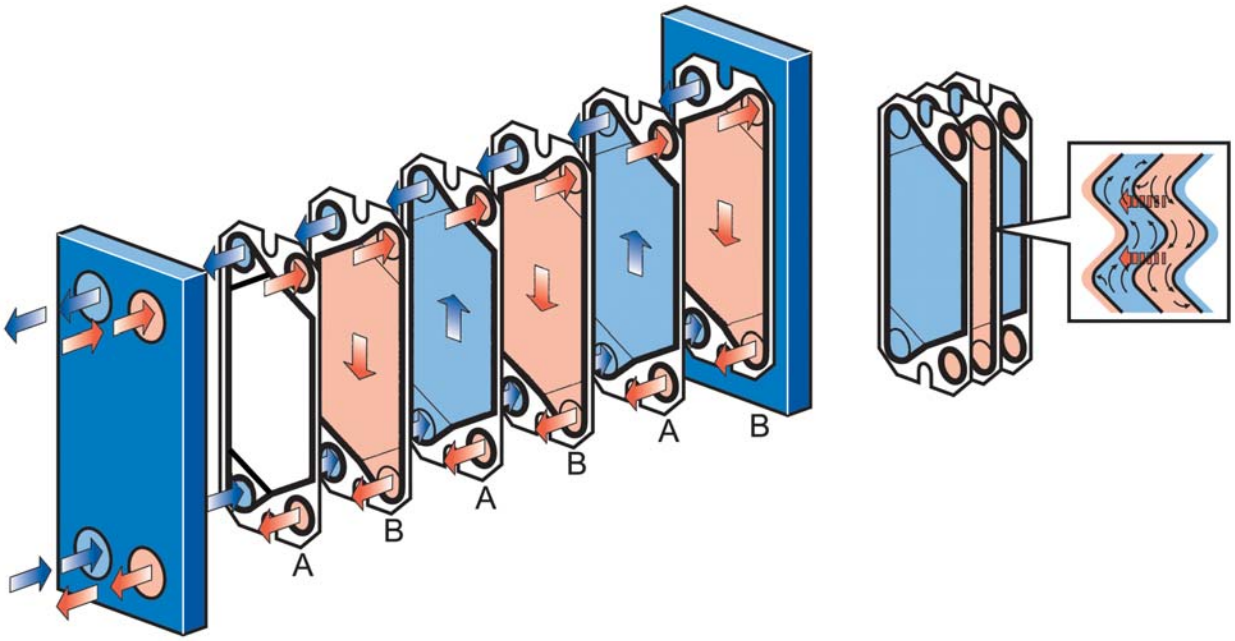
Bolt protection	Plastic tubes that protect the threads of the tightening bolts.
Carrying bar	Carries the plate pack and the pressure plate.
Frame plate	Fixed steel plate with a number of port holes for the connection of the piping system. The carrying and guiding bars are supported by the frame plate.
Guiding bar	Keeps the channel plates and the pressure plate in line at their lower end.
Plate pack	Heat is transferred from one medium to the other through the plates. The plate pack consists of channel plates, end plates, gaskets and, in some cases, transition plates. The measure of the plate pack is the A dimension, i.e the measurement between frame and pressure plate. Refer to PHE drawing.
Port holes with stud bolt connections	Port holes through the frame plate allow the media to enter into or exit from the Plate Heat Exchanger. Different types of connections can be used to connect the piping system to the apparatus. Threaded stud bolts around the port holes secure the connections to the apparatus. The port holes may be protected against corrosion by metal or rubber linings. The PHE can be equipped with different connection types. For details refer to PHE drawings.
Pressure plate	Moveable steel plate that can contain a number of port holes for the connection of the piping system.
Protective sheets	Cover the plate pack. Mandatory in the USA. Optional in other countries.
Supporting column	Supports carrying and guiding bars.
Tightening bolts	Compress the plate pack between the frame and pressure plate.

EN

Function

The Plate Heat Exchanger (PHE) consists of a pack of corrugated metal plates with port holes for input and output for the two separate fluids. The heat transfer between the two fluids will take place through the plates.

The plate pack is assembled between a frame plate and a pressure plate and compressed by tightening bolts. The plates are fitted with a gasket that seals the channel and directs the fluids into alternate channels. The plate corrugation promotes fluid turbulence and supports the plates against differential pressure.

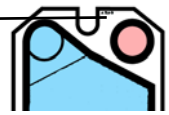


Principle of plate pack arrangement

Identification of plate side

The A side of the plate is identified by the stamp with the **letter A** or **the model name,** in some cases both, at the top of the plate (see figure on the right-hand side).

Identification stamp



Semi-welded PHE

For certain plate sizes there are semi-welded plates (cassettes) available. The function of the semi-welded PHE is the same as that of the conventional PHE. The chapters Installation and Operation in this manual are fully applicable here.

The chapter Maintenance is fully applicable when it comes to the parts Cleaning-In-Place and Pressure test after maintenance and partially applicable for the remaining part.

EN

Installation

Before installation

To consider before installation

- Before connecting any piping, make sure all foreign objects have been flushed out of the piping system that should be connected to the PHE.
- Before start-up, check that all tightening bolts are firmly tightened and that the correct measurements of the plate pack are used. Refer to PHE drawing.
- When connecting the piping system make sure the pipes do not subject the PHE to stress or strain.
- To avoid water hammer, do not use fast-closing valves.
- Safety valves should be installed according to current pressure vessel regulations.
- If the PHE surface temperature is expected to be hot or cold, the PHE should be insulated.
- It is recommended that protective sheets are used to cover the plate pack.
- For each model, design pressures and temperatures are marked on the identification plate. These must not be exceeded.

Requirements

Space

A minimum free space is needed for lifting plates in and out. Refer to the delivered drawing.

Foundation

Install on a flat foundation giving enough support to the frame.

Elbow

To make it easier to disconnect the PHE, an elbow should be flanged to the connection in the pressure plate, directed upwards or sideways, and with another flange located just outside the contour of the Plate Heat Exchanger.

Shut-off valve

To be able to open the PHE, shut-off valves should be provided in all connections.

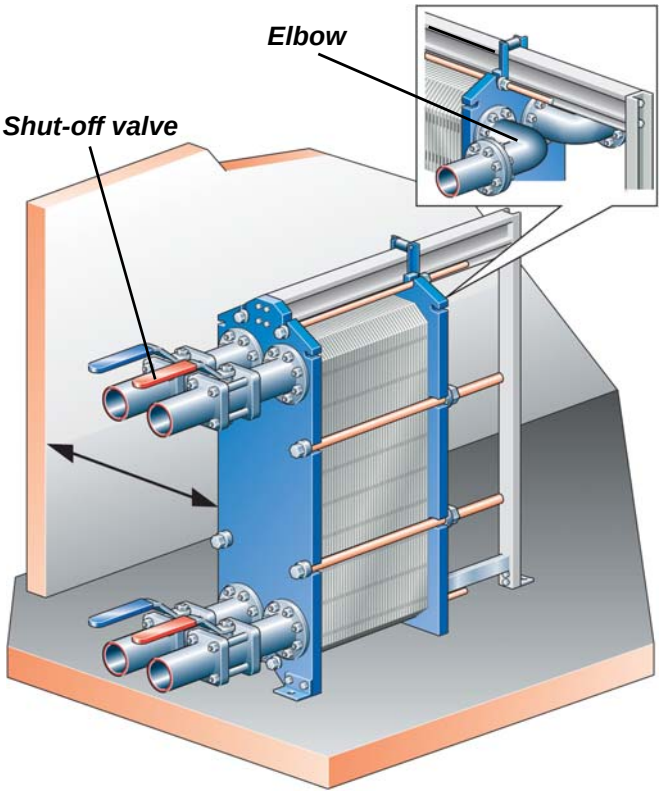
Drip tray (optional)

Depending on the type of fluid in the PHE and the type of installation, a drip tray (drainage box) may be necessary to avoid injury to personnel and damage to equipment.

Note!
Put the drip tray in place before positioning the PHE.

Connections in the pressure plate

It is important that the plate pack has been tightened to the correct dimension A (check against PHE drawing) before the piping system is connected.

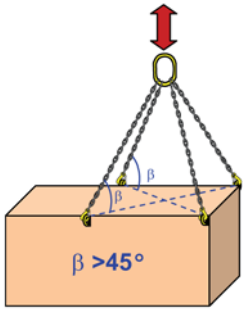


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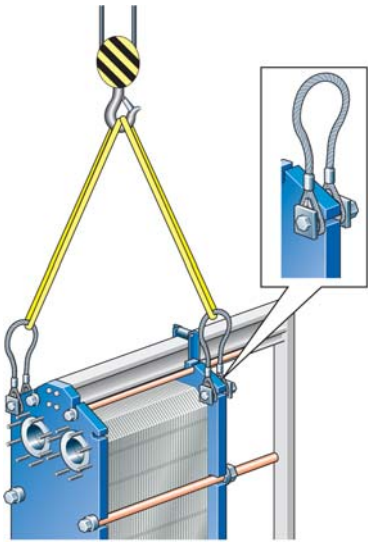
Lifting

Straps should be used when lifting the PHE. Place straps according to picture.

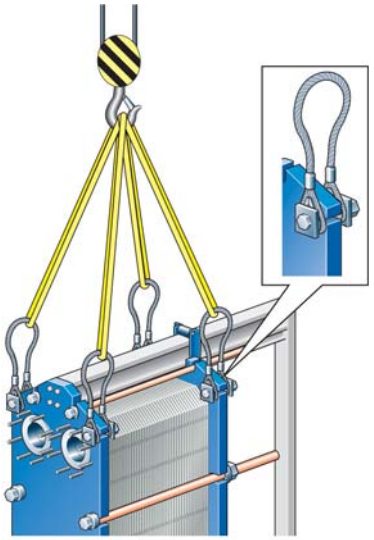
The straps shall be attached according to the figures and with a minimum angle of 45°. Refer to the figure below.



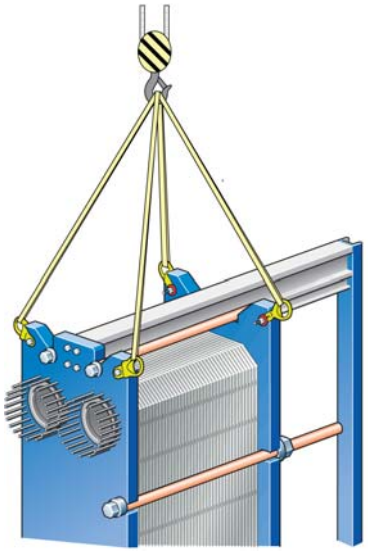
Warning!
Never lift by the connections or the studs around them.



Lifting device for M15, TL10, TL15, T20, TS20.



Lifting device for MX25, M30 and MA30.



Lifting device for TL35.

For detailed information, contact an Alfa Laval Sales Representative for the document “Cargo Securing Instructions” (3490003791, 3490003792 and 3490003793).

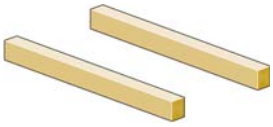
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Raising

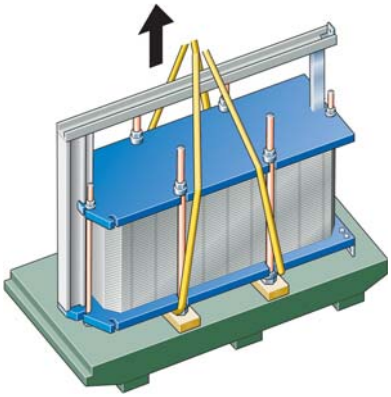
This instruction is valid when raising the PHE after delivery from Alfa Laval. Only use a strap approved for the weight of the PHE.

Caution!
The straps shall be long enough to be able to rotate the PHE without obstructions. Especially consider the space for the support column.

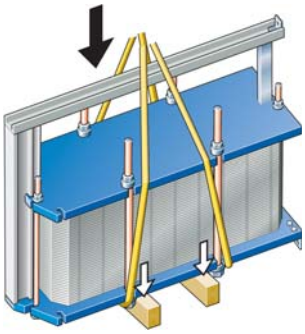
1 Place two timber beams on the floor.



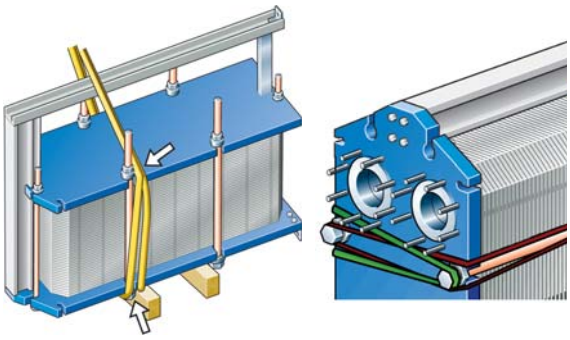
2 Lift the PHE off the pallet using e.g. straps.



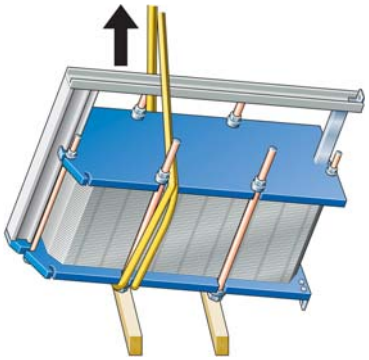
3 Place the PHE on the timber beams.



4 Place straps around one bolt on each side.



5 Lift the PHE off the timber beams.



6 Lower the PHE into a horizontal position and place it on the floor.



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Operation

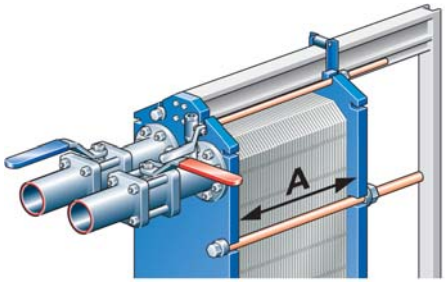
Start-up

During start-up, check that no visible leakages appear from the plate pack, valves or piping system.

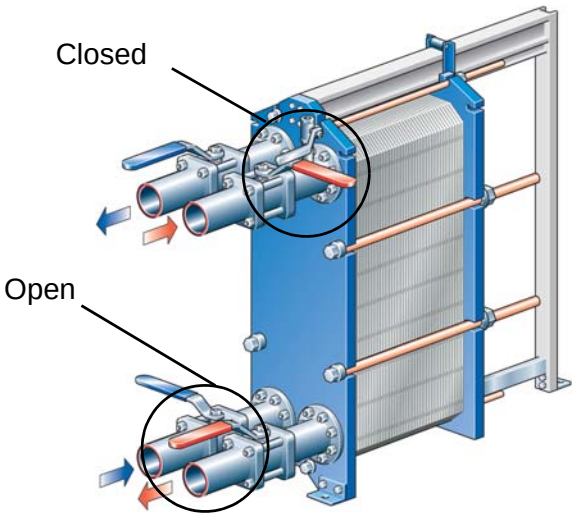
Note!
If several pumps are included in the system, make sure you know which one should be activated first.

Note!
Adjustments of flow rates should be made slowly in order to avoid the risk of **pressure surge (water hammer)**.
Water hammer is a short-lasting pressure peak that can appear during start-up, or shut-down of a system, causing liquids to travel along a pipe as a wave at the speed of sound. This can cause considerable damage to the equipment.

1 Before start-up check that all tightening bolts are firmly tightened and that the dimension A is correct. Refer to PHE drawing.

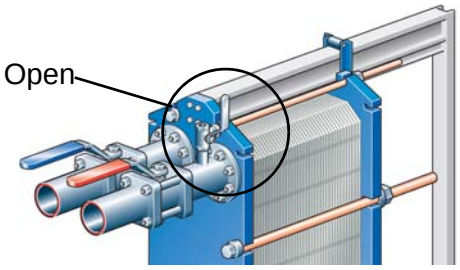


2 Check that the valve is closed between the pump and the unit controlling the system flow rate.

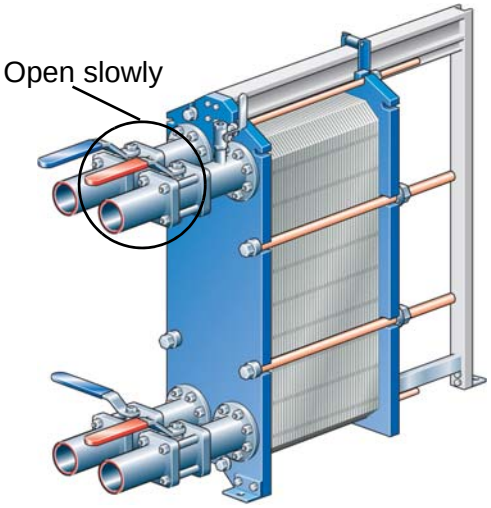


3 If there is a valve at the exit, make sure it is fully open.

4 Open the air vent and start the pump.

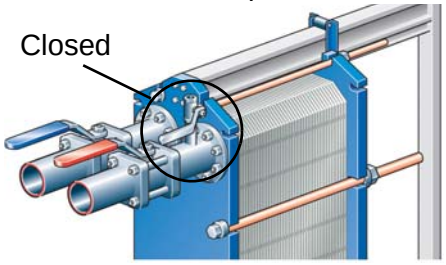


5 Open the valve slowly.



Note!
Avoid rapid temperature changes in the PHE. With media temperatures over 100°C, slowly increase the temperature preferably at least for one hour.

6 When all air is expelled, close the air vent.



7 Repeat steps 1– 6 for the second media.

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Unit in operation

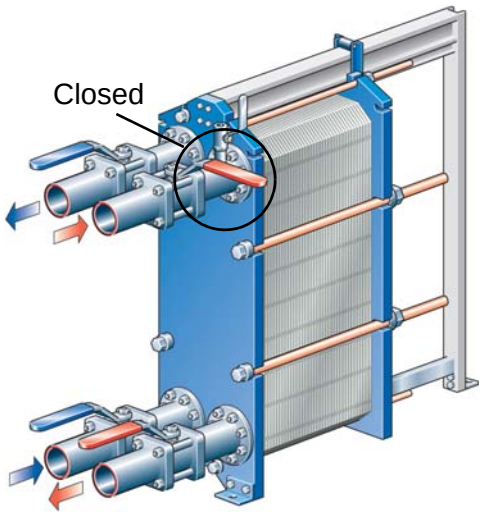
Adjustments of flow rates should be made slowly in order to protect the system against sudden and extreme variations of temperature and pressure.

During operation, check that media temperatures and pressures are within the limits stated on the PHE-drawing and identification plate.

Shut-down

Note!
If several pumps are included in the system, make sure you know which one should be stopped first.

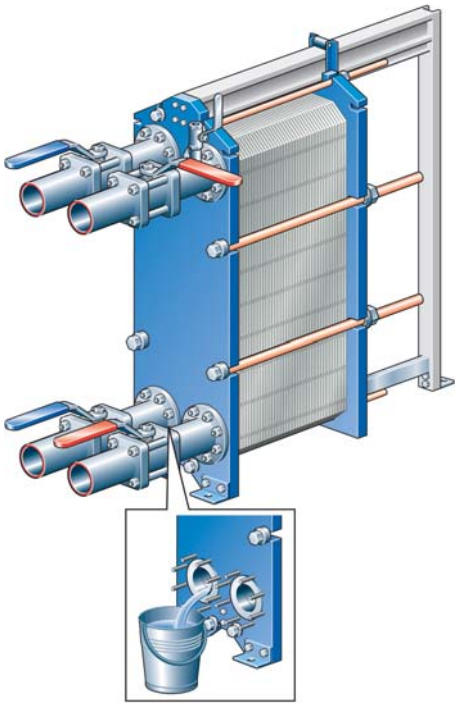
1 Slowly close the valve controlling the flow rate of the pump you are about to stop.



2 When the valve is closed, stop the pump.

3 Repeat steps 1–2 for the other side for the second media.

4 If the PHE is shut down for several days or longer, it should be drained. Draining should also be done if the process is shut down and the ambient temperature is below the freezing temperature of the media. Depending on the media processed, it is also recommended to rinse and dry the PHE plates and connections.



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Maintenance

To keep the Plate Heat Exchanger in good condition regular maintenance is required.

The plates need to be cleaned on a regular basis. The frequency depends on several factors such as type of media and temperatures. Three methods can be used to perform cleaning. Refer to “Cleaning-In-Place (CIP)” on page 12, “Manual cleaning” on page 13 or a reconditioning at an Alfa Laval service center.

After a longer period of use, it may be required to re-gasket the PHE by exchanging the gaskets. Refer to “Regasketing” on page 19.

Other maintenance that should be performed regularly:

- Keep carrying bar and guiding bar cleaned and greased
- Keep the tightening bolts cleaned and greased.

Cleaning-In-Place (CIP)

The Cleaning-In-Place (CIP) equipment permits cleaning of the PHE without opening it. The results of cleaning with CIP are as follows:

- Cleaning of fouling and descaling of lime deposits
- Passivation of cleaned surfaces to reduce susceptibility to corrosion
- Neutralization of cleaning liquids before draining.

Follow the instructions of the CIP equipment.

Warning!
Use proper protective equipment, such as safety boots, safety gloves and eye protection, when using the cleaning agents.



Warning!
Corrosive cleaning liquids can cause serious injuries to skin and eyes!



Alfa Laval guarantees that plates, gaskets or glue are not damaged if the procedures given and cleaning agents prescribed are followed.

If CIP cannot be done, cleaning must be done manually. Refer to section “Manual cleaning” on page 13.



CIP equipment

Model	***Maximum number of plates***			
	CIP 200	***CIP 400***	***CIP 800***	***CIP 1800***
M15-B/BD/E/M	316	600	600	600
TL10-B/P	333	522	522	522
TL15-B	165	257	665	665
T20-B/M/P	160	348	700	700
TS20-M	176	382	841	1971
MX25-B/M	83	180	395	925
M30	48	103	227	532
MA30-M/S/SM	49	107	234	549
TL35-B	55	120	263	617

Cleaning liquids

Liquids	***Description***
AlfaCaus	A strong alkaline liquid, for removing paint, fat, oil and biological deposits.
AlfaPhos	An acid cleaning liquid for removing metallic oxides, rust, lime and other inorganic scale. Contains repassivation inhibitor.
AlfaNeutra	A strong alkaline liquid for the neutralization of AlfaPhos before drainage.
Alfa P-Neutra	For the neutralization of Alfa P-Scale.
Alfa P-Scale	An acidic powder cleaner for the removal of of primary carbonate scale but also other inorganic scale.
AlfaDescalent	A non-hazardous acidic cleaning agent for the removal of inorganic scale.
AlfaDegreaser	A non-hazardous cleaning agent for the removal of oil, grease or wax deposits. Additionally prevents foaming when using Alpacon Descaler.

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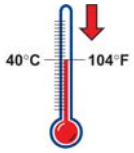
Manual cleaning

To perform manual cleaning it is required to open the PHE and lift out the plates to clean them.

Opening

Note!
Before opening the PHE check the warranty conditions. If in any doubt, contact an Alfa Laval sales representative. Refer to "Warranty conditions" on page 2.

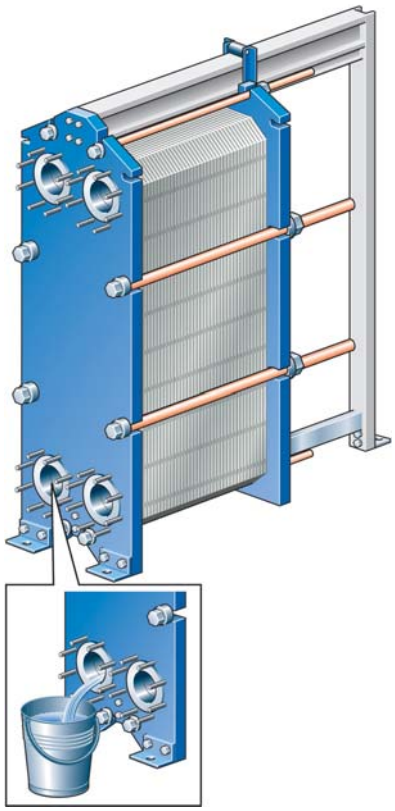
Warning!
If the Plate Heat Exchanger is hot, wait until it has cooled down to about 40°C (104°F).



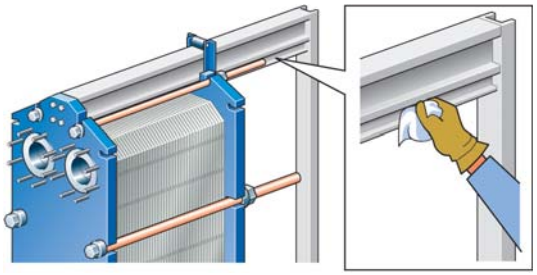
Warning!
If necessary, use proper protective equipment, such as safety boots, safety gloves and eye protection, depending on type of media in the PHE.



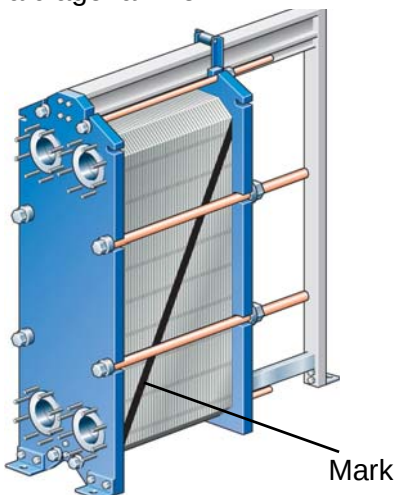
1 Drain the Plate Heat Exchanger.



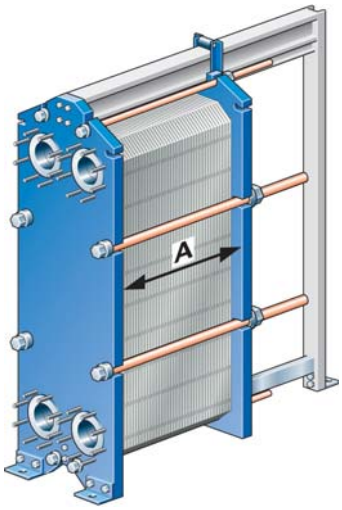
2 Inspect the sliding surfaces of the carrying bar and clean and grease it.



3 Mark the plate assembly on the outside with a diagonal line.



4 Measure and note the dimension A.

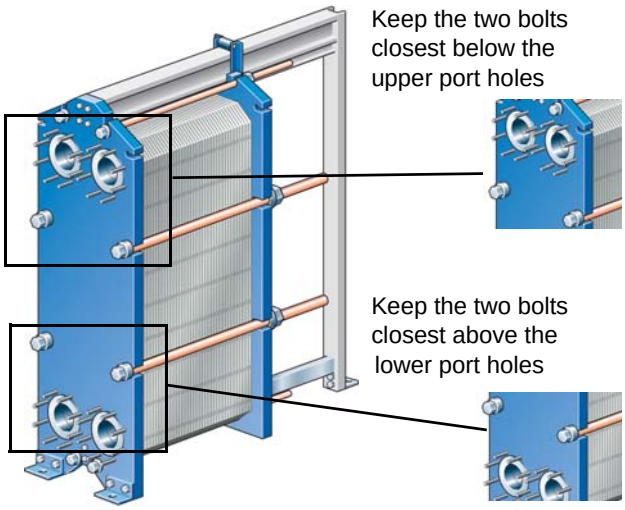


Note!
Brush the threads of the tightening bolts with a steel wire brush and then grease before loosening them.

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5

Keep the four bolts in position, according to the figure below. Loosen the other bolts and remove them.



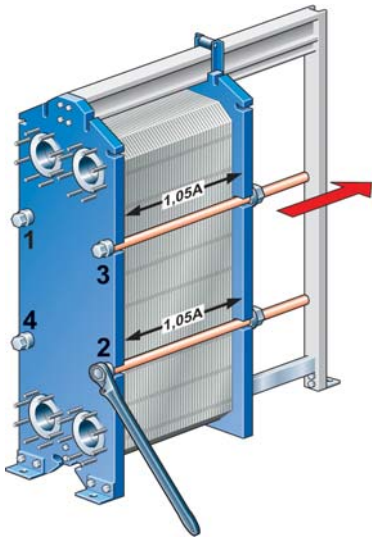
6

The remaining four bolts are opened alternately and diagonally in two steps, see figures below.

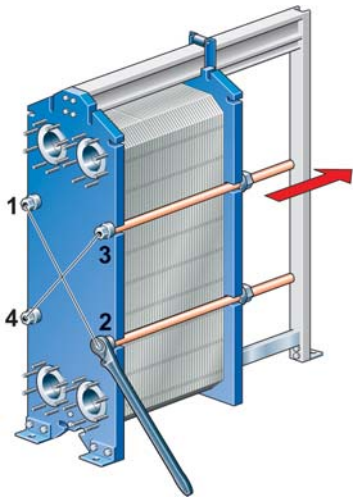
Be careful to ensure that the frame plate and pressure plate are always in parallel. Skewing of the pressure plate during opening must not exceed 10 mm (**2 turns per bolt**) across the width and 25 mm (**5 turns per bolt**) vertically.

Step	*Bolt No.*	*To dimension*
1	1–2–3–4	1.05A
2	1–2 or 3–4	Opening

Step 1: Loosen the four bolts alternately and diagonally until the plate package measures 1.05A.



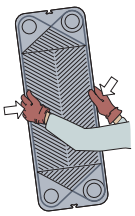
Step 2: Loosen the two diagonal pairs of bolts alternately, as shown in the figure below.



7

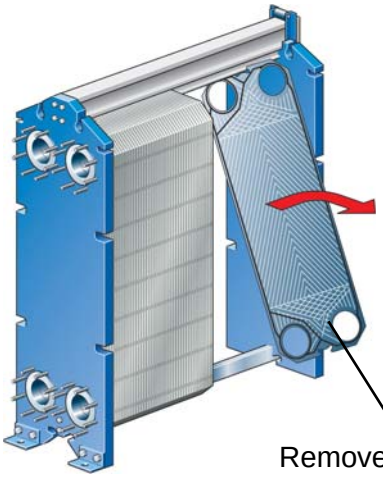
Open the plate pack by letting the pressure plate glide on the carrying bar.

Caution!
To avoid hand injuries from sharp edges, protective gloves should always be worn when handling plates and protective sheets.



Note!
Plates should be numbered, do this before removing the plates.

Plates do not need to be removed if cleaning is done using only water, i.e. without cleaning agent.



Warning!
The plate pack may still contain a small residual amount of liquid after draining. Depending on the type of product and type of installation, special arrangements, e.g. drainage box, may be necessary to avoid injury to personnel and damage to equipment.

EN

Manual cleaning of opened units

Caution!
Never use hydrochloric acid with stainless steel plates. Water of more than 330 ppm Cl may not be used for the preparation of cleaning solutions. It is very important that carrying bars and support columns in aluminium are protected against chemicals.

Note!
Be careful not to damage the gasket during manual cleaning.

Deposits removable with water and brush

Plates do not need to be removed from the PHE during cleaning.

Warning!
If necessary, use proper protective equipment. Consider risks like loose particles and what kind of media has been used in the PHE.

1 Start cleaning when the heating surface is still wet and the plates are hanging in the frame.

2 Remove deposits using a soft brush and running water.



3 Rinse with water using a high pressure hose.



Deposits not removable with water and brush

Plates must be removed from the PHE during cleaning. For a choice of cleaning agents, refer to "Cleaning liquids" on page 12.

Warning!
Use proper protective equipment, such as safety boots, safety gloves and eye protection, when using the cleaning agents.



Warning!
Corrosive cleaning liquids can cause serious injuries to skin and eyes!



1 Brush with cleaning agent.



2 Rinse immediately with water.



Note!
Long exposure to the cleaning agents can damage the gasket glue.

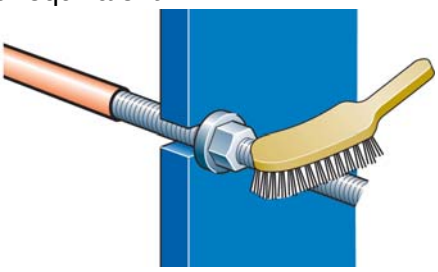
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Closing

Follow the instructions below to ensure that the Plate Heat Exchanger will be properly closed. For TL15-B, follow the separate instructions, Refer to “Closing - TL15-B” on page 18.

1 Check that all the sealing surfaces are clean.

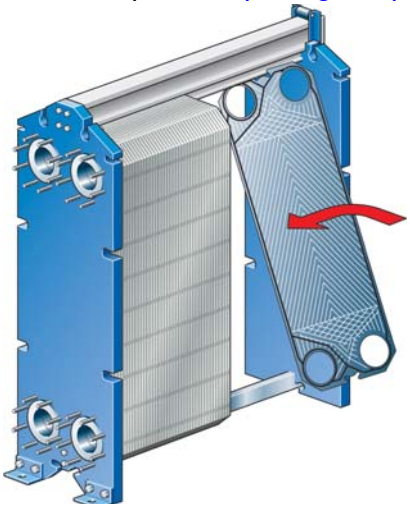
2 Brush the threads of the bolts clean, using a steel wire brush or the Alfa Laval thread cleaner. Lubricate the threads with a thin layer of grease, e.g. Gleitmo 800 Lubriplate or equivalent.



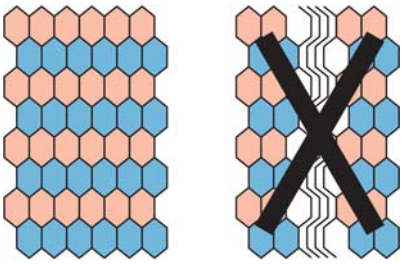
3 Attach gaskets to the plates or check that all the gaskets are properly attached.

> ***Note!***
> If the gasket is wrongly positioned, it will show by the fact that it rises out of the gasket groove or that it is positioned outside the groove.

4 Insert the plates in alternate directions and with the gaskets turned towards the frame plate or pressure plate as specified on the plate hanging list. Use the marked line that was done when the PHE was opened. Refer to step 3 in “Opening” on page 13.



5 If the plates are correctly assembled, the edges form a “honeycomb” pattern, see picture below.

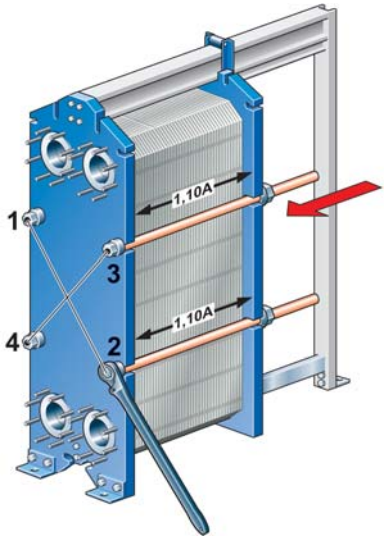


6 Press the plate assembly together. Tightening is done in two steps, see figures below. Be careful to ensure that the frame plate and the pressure plate are always in parallel.

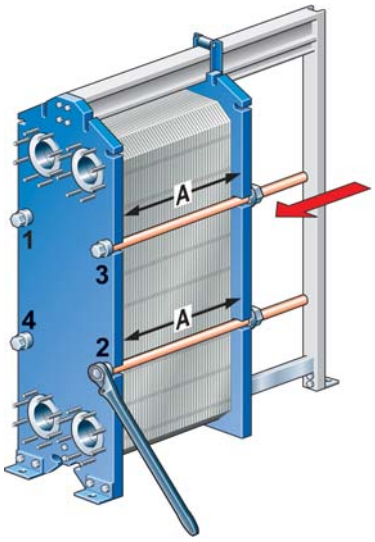
Step	*Bolt No.*	*To dimension*
1	1–2 or 3–4	1.10A
2	1–2–3–4	A

Step 1: Tighten the two diagonal pairs of bolts alternately until the plate package measures 1.10A.

Be careful to ensure that the frame plate and pressure plate are always in parallel. Skewing of the pressure plate during opening must not exceed 10 mm (**2 turns per bolt**) across the width and 25 mm (**5 turns per bolt**) vertically.



Step 2: After that, bolts are tightened alternately and diagonally, as shown in the figure below. Check the dimension A during tightening at the positions of the bolts that are being used.



Note!
The final tightening to reach dimension A is recommended to be divided into steps.

Max tightening torque

When a pneumatic tightening device is used, see table below for maximum torque. Measure dimension A during tightening.

Bolt size	***Bolt with bearing box***		***Bolt with washers***	
	N·m	kpm	N·m	kpm
M24			450	45
M30			900	90
M39	1300	130	2000	200
M48	2100	210	3300	330

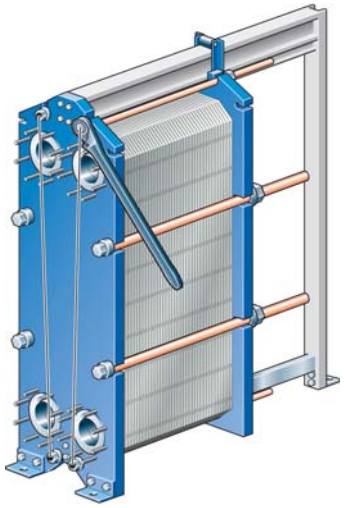
For manual tightening, the tightening torque has to be estimated.

If dimension A cannot be reached

- Check the number of plates and the dimension A.
- Check that all the nuts and bearing boxes are running freely. If not, clean and lubricate, or replace.

7 Place the other bolts in position.

- Inspect the washers.
- When fully tightened, the bolts should all be equally tensioned.



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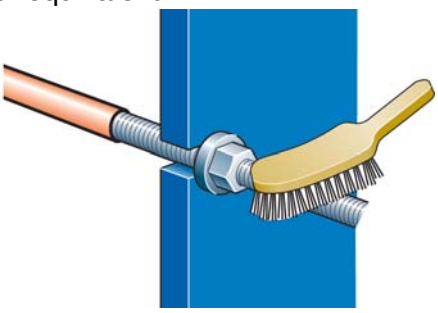
Closing - TL15-B

Follow the instructions below to ensure that the Plate Heat Exchanger will be properly closed.

Closing instructions

1 Check that all the sealing surfaces are clean.

2 Brush the threads of the bolts clean, using a steel wire brush or the Alfa Laval thread cleaner. Lubricate the threads with a thin layer of grease, e.g. Gleitmo 800 Lubriplate or equivalent.



3 Attach gaskets to the plates or check that all the gaskets are properly attached.

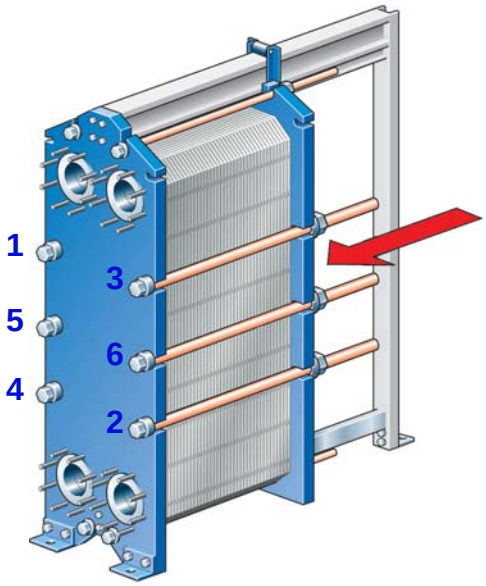
Note!
If the gasket is wrongly positioned, it will show by the fact that it rises out of the gasket groove or that it is positioned outside the groove.

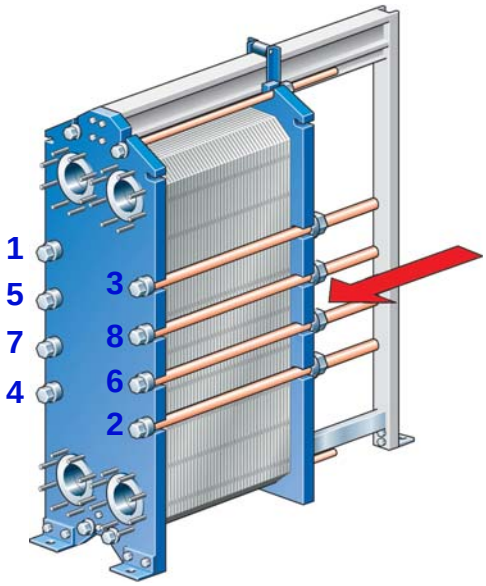
4 Insert the plates in alternate directions and with the gaskets turned towards the frame plate or pressure plate as specified on the plate hanging list. Use the marked line that was done when the PHE was opened. Refer to step 3 in “Opening” on page 13.

5 Press the plate package together and put the tightening bolts in their positions.

6 Tighten the bolts alternately in numeric order, 1 to 6 (8). Repeat this procedure until the dimension A is reached.

Note!
Make sure that the frame plate and the pressure plate are always positioned in parallel (within maximum 10 mm).





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Pressure test after maintenance

Before start-up of production, whenever plates or gaskets have been removed, inserted or exchanged, it is strongly recommended to perform a pressure test to confirm the internal and external sealing function of the PHE. During this test, one media side at a time must be tested with the other side open to the ambient pressure.

Caution!
The pressure testing shall be performed at a pressure equal to the operating pressure of the actual unit, but never above the design pressure as stated on the nameplate.

The recommended test time is 10 minutes for each media.

Please note that PHE units for refrigeration applications and units with media not mixable with water must be dried after hydrostatic pressure testing.

Please consult the local office/representative of the supplier for advice on the pressure testing procedure.

Regasketing

The procedures below relate to Field gaskets, Ring gaskets and End gaskets.

Note!
Before removing the old gaskets check how they are attached.

Clip-on

1 Open the PHE, [refer to “Opening” on page 13] and remove the plate that is to have a new gasket.

Note!
Before opening the PHE check the warranty conditions. If in any doubt, contact an Alfa Laval sales representative. Refer to “Warranty conditions” on page 2.

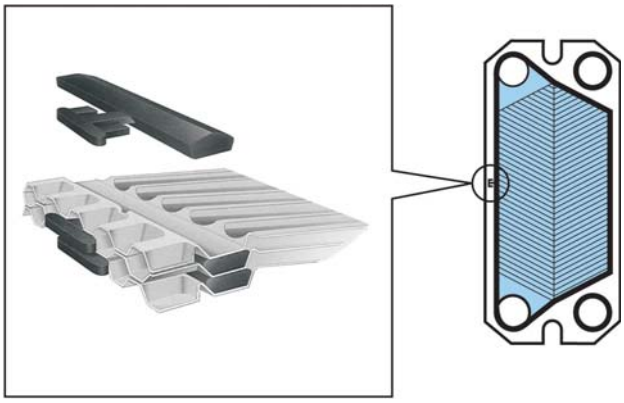
2 Remove the old gasket.

3 Ensure that all sealing surfaces are dry, clean and free of foreign matter such as fat, grease or similar.

4 Check the gasket and remove rubber residual before attaching it.

Note!
Especially the end plate gasket!

5 Attach the clip-on gasket to the plate. Slip the gasket prongs under the edge of the plate.



Note!
Make sure the two gasket prongs are in the correct position.

6 Repeat the procedure until all plates that are needed to be regasketed are done. Close the PHE according to “Closing” on page 16.

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Clip-AD gaskets (MX25 and TL15)

The Clip-AD gasket represents a system with the conventional Clip-on fastening around the ports and fastening by means of adhesive tape along the sides of the plates.

The use of the adhesive tape (GC1) is a simple way to obtain secure gasket positioning. It is adhered to the gasket groove by means of a special tape gun, making it easy to apply the tape exactly where wanted.

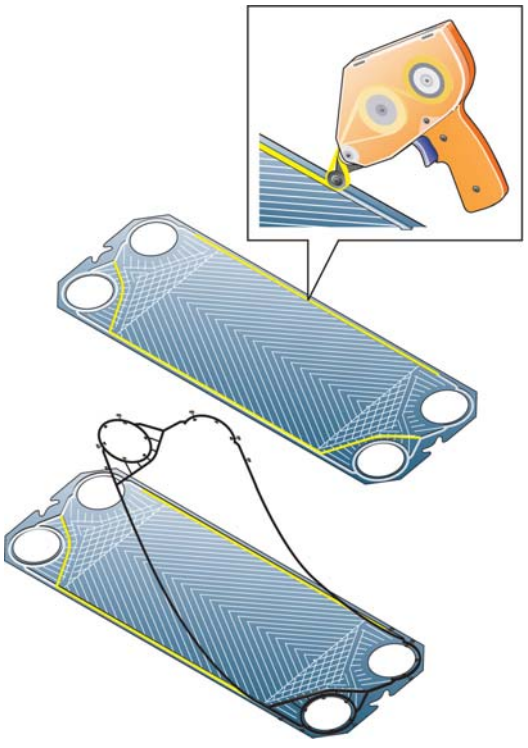
1 Open the PHE [refer to “Opening” on page 13] and remove the plate that is to have a new gasket.

Note!
Before opening the PHE check the warranty conditions. If in any doubt, contact an Alfa Laval sales representative. Refer to “Warranty conditions” on page 2.

2 Remove the old gasket.

3 It is not necessary to remove old tape as the film is very thin. Make sure, however, that the gasket groove is clean and dry.

4 Adhere tape, using the tape gun.



5 Attach the gasket to the plate. Slip the gasket prongs under the edge of the plate.

6 Close the plate heat exchanger according to “Closing” on page 16.

Glued gaskets

Use glue recommended by Alfa Laval. Separate gluing instructions will be delivered together with the glue.

Caution!
Other glues than those recommended can contain chlorides that can damage the plates.

Caution!
Do not use sharp tools when removing the glued gasket to avoid damage to the plates.

Semi-welded gaskets

These instructions are is only valid when cassettes are used. Refer to Plate hanging list!

1 Follow steps 1 to 4 in the instructions for Clip-on gaskets. Refer to “Clip-on” on page 19.

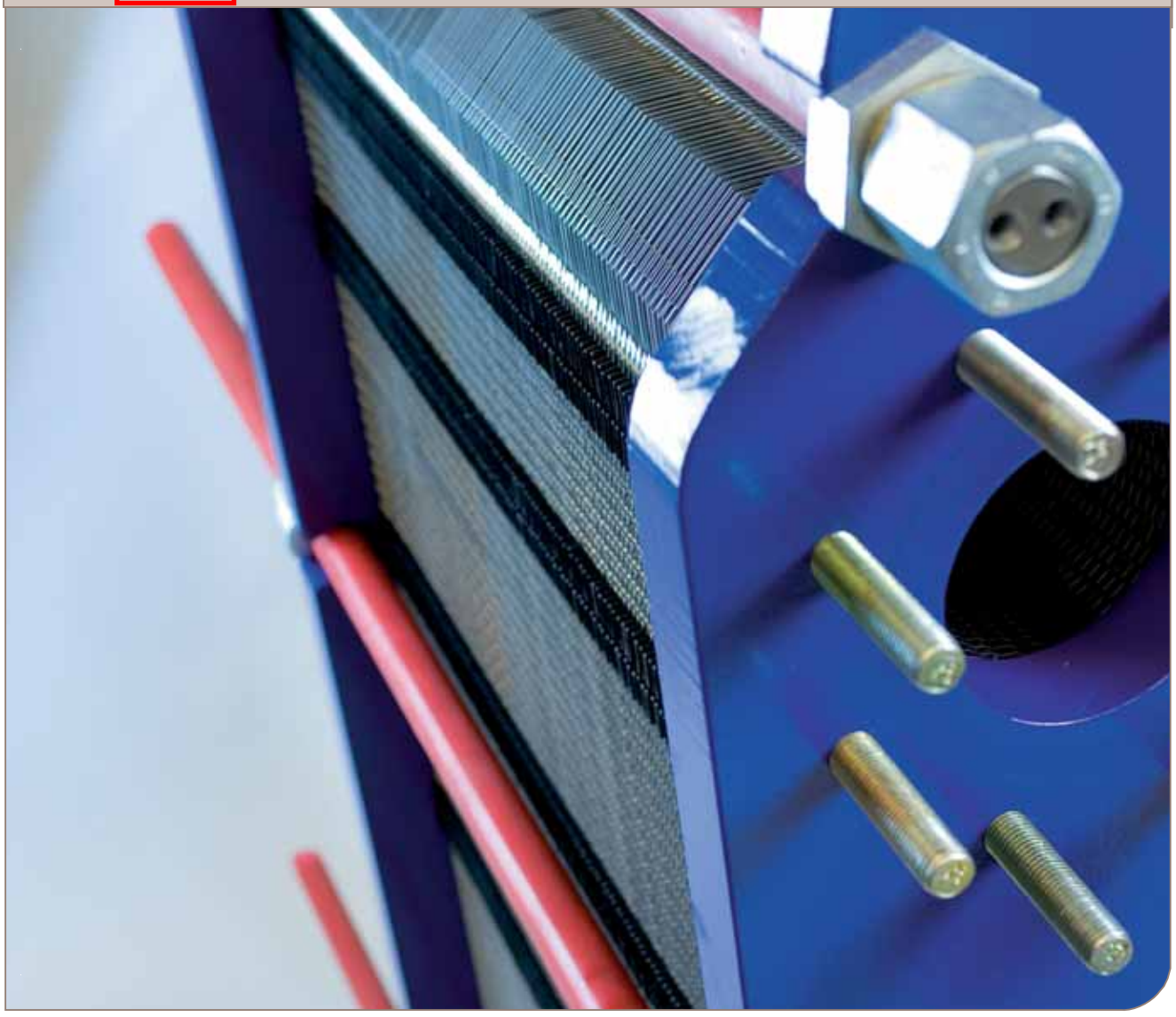
2 Channel Cassette - Attach adhesive tape to the gasket groove.

3 Place the Ring gasket and attach it with the clip-on.

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Instruction Manual - Plate Heat Exchangers

M3, M6, M10, TS6, T2, T5, TL3, TL6



3350012572 2012-09

Original manual

Table of contents

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Description 6
- Components 6
- Name Plate 8
- Function 9
- Multi-pass 10
- Identification of plate side 10

Installation 11
- Before installation 11
- Requirements 11
- Lifting 13
- Raising 13

Operation 15
- Start-up 15
- Unit in operation 17
- Shut-down 17

Maintenance 18
- Cleaning – Non-product side 18
- Opening 20
- Manual cleaning of opened units 24
- Closing 26
- Pressure test after maintenance 28
- Regasketing 29

How to contact Alfa Laval
Contact details for all countries are
continually updated on our website.
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Preface

This manual provides information needed to install, operate and carry out maintenance of the Plate Heat Exchanger (PHE).

Safety considerations

The Plate Heat Exchanger shall be used and maintained in accordance with Alfa Laval's instructions in this manual. Faulty handling of the Plate Heat Exchanger may result in serious consequences with injuries on persons and/or property damage. Alfa Laval will not accept responsibility for any damage or injury resulted from not following the instructions in this manual.

The Plate Heat Exchanger shall be used in accordance with the specified configuration of material, media types, temperatures and pressure for the specific Plate Heat Exchanger.

The following models are covered in this manual:

- M3
- M6
- M10
- TS6
- T2
- T5
- TL3
- TL6

Prior knowledge

The PHE shall be operated by persons that have studied the instructions in this manual and have knowledge of the process. It also considers knowledge of precautions regarding media type, pressures, temperatures in the PHE as well as specific precautions required of the process.

Maintenance and installation of the PHE shall be done by persons that have knowledge and authorization according to the local regulations. This may include actions such as piping, welding and other kind of maintenance.

Maintenance actions not described in this manual please contact Alfa Laval representative for advice.

Definitions of expressions

Warning!
Type of hazard
WARNING indicates a potentially hazardous situation that, if not avoided, could result in death or serious injury.

Caution!
Type of hazard
CAUTION indicates a potentially hazardous situation that, if not avoided, may result in minor or moderate injury.

Note!
NOTE indicates a potentially hazardous situation that, if not avoided, may result in property damage.

PHE drawings

PHE drawings mentioned in the manual is the drawings included in the delivery of the Plate Heat Exchanger.

Warranty conditions

The warranty conditions are usually included in the signed sales contract prior to the order of the delivered PHE. Alternatively the warranty conditions is included in the sales offer documentation or with a reference to the document specifying the valid conditions. If faults occur during the specified warranty period always consult your local Alfa Laval Representative for advice.

Report the date when the Plate Heat Exchanger has been taken in operation to the local Alfa Laval Representative.

Advice

Always consult your local Alfa Laval Representative for advice on

- New plate pack dimensions if you intend to change number of plates
- Selection of gasket material if operating temperatures and pressures are permanently changed, or if another medium is to be processed in the PHE.

Storage of the PHE

Alfa Laval delivers the PHE ready to be put in service upon arrival, if nothing else has been agreed. Although keep the PHE in the packing box until the installation.

Storage for longer periods of time, one month or longer, certain precautions should made to avoid unnecessary damages on the PHE.

Note!
Alfa Laval and its representatives reserve the right to inspect the storage space and /or equipment whenever necessary until expiry of the warranty period stipulated in the contract. Notification has to be given 10 days prior to date of inspection.

If any uncertainty appear about storage of the PHE consult an Alfa Laval Representative.

Storage in packing box

If storage after delivery of the PHE is known in advance, inform Alfa Laval when ordering the PHE to ensure that it will be properly prepared for storage before packing.

Indoor storage

- Store inside a room with the temperature between 15 and 20°C (60 - 70°F) and humidity around 70%. For outdoor storage read “Outdoor storage” on this page.
- To prevent damage to the gaskets, there should not be any ozone-producing equipment in the room such as electric motors or welding equipment.
- To prevent damage to the gaskets, do not store organic solvents or acids in the room and avoid direct sunlight, intensive heat radiation or ultraviolet radiation.
- The tightening bolts should be well covered with light grease coating.

Outdoor storage

If the PHE has to be stored outdoors, all the precautions mentioned in the section “Indoor storage” on this page should be taken. Also, protection against climate is very important.

The stored PHE shall be visually checked every third month. The check includes:

- Greasing of the tightening bolts
- Metal port covers
- Protection of the plate pack and gaskets

Taken out of service

If, for any reason, the PHE is shut down and taken out of service for a long period of time, follow the same advice as in the previous section ”Indoor storage” on this page. Although before storage following actions has to be done.

- Check the measurement of the plate pack (measure between frame and pressure plate, A dimension).
- Drain both media sides of the PHE
- Depending on the media, the PHE should be rinsed and then dried.
- The connection should be covered if the piping system is not connected. Use a plastic or plywood cover for the connection.
- Cover the plate pack with non-transparent plastic film.

Installation after long-term storage

In cases when the PHE has been taken out of service for an extensive period of time, i.e. longer than one year, the risk of leakage when starting up increases. To avoid this problem it is recommended to let the gasket rubber rest and regain most of its elasticity.

1. If the PHE is not in position, follow the instructions “Installation” on page 11.
2. Note the measurement between frame and pressure plate (A dimension).
3. Remove feet attached to the pressure plate.
4. Loosen the tightening bolts. Follow the instructions “Opening” on page 20. Open the PHE until the measure is 1.25A.
5. Leave the PHE between 24-48 hours, the longer the better, for gaskets to relax.
6. Re-tighten according to the instructions “Closing” on page 26.
7. Alfa Laval recommends a hydraulic test to be carried out. The media, usually water, should be entered at intervals to avoid sudden shocks to the PHE. It is recommended to test up to the Design Pressure, refer to PHE drawing.

Environmental compliance

Alfa Laval endeavours to perform its own operations as cleanly and efficiently as possible, and to take environmental aspects into consideration when developing, designing, manufacturing, servicing and marketing its products.

Unpacking

Packing material consists of wood, plastics, cardboard boxes and, in some cases, metal straps.

- Wood and cardboard boxes can be reused, recycled or used for energy recovery.
- Plastics should be recycled or burnt at a licensed waste incineration plant.
- Metal straps should be sent for material recycling.

Maintenance

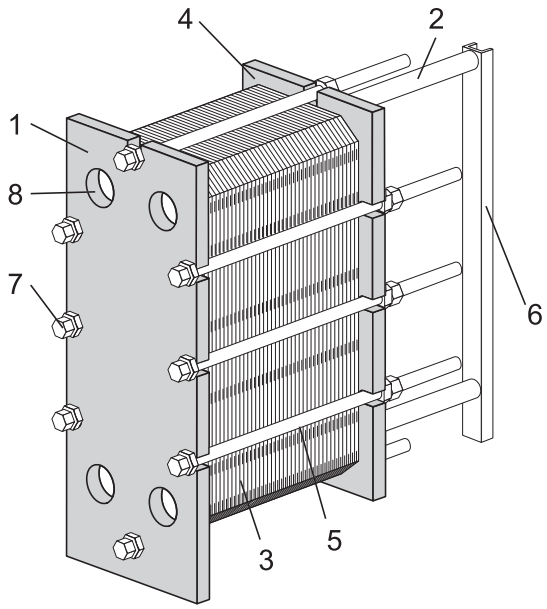
- All metal parts should be sent for material recycling.
- Oil and all non-metal wear parts must be taken care of in agreement with local regulations.

Scrapping

At end of use, the equipment shall be recycled according to relevant, local regulations. Besides the equipment itself, any hazardous residues from the process liquid must be considered and dealt with in a proper manner. When in doubt, or in absence of local regulations, please contact the local Alfa Laval sales company.

Description

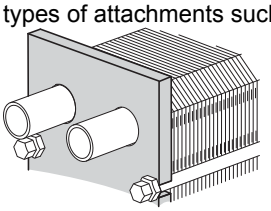
Components



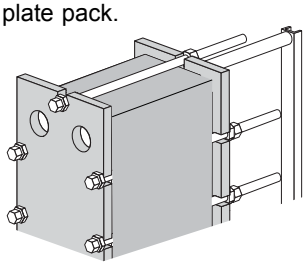
Main components

1	***Frame plate***	Fixed plate with various number of port holes for connection of the piping system. The carrying and guiding bar is attached to the frame plate.
2	***Carrying bar***	Carries the plate pack and the pressure plate.
3	***Plate pack***	Heat is transferred from one media to the other through the plates. The plate pack consists of channel plates, end plates, gaskets and in some cases transition plates. The measure of the plate pack is the A dimension, i.e the measurement between frame and pressure plate. Refer to PHE drawing.
4	***Pressure plate***	Moveable plate that can contain various number of port holes for connection of the piping system.
5	***Guiding bar***	Keeps the channel plates, connection plates and the pressure plate alligned at their lower end.
6	***Supporting column***	Supports carrying and guiding bars. For some smaller PHE models no supporting column is used.
7	***Tightening bolts***	Compress the plate pack between the frame and pressure plate. There are usually four tightening bolts used, in some cases six, these are used to open and close the PHE. Remaining bolts are used as locking bolts.
8	***Port holes***	Port holes through the frame plate allows the media to enter into or exit from the Plate Heat Exchanger. Different types of connections can be used to connect the piping system to the apparatus. The port holes may be protected against corrosion by metal or rubber linings.

Connections

Pipe connections	The Plate Heat Exchanger can be equipped with of fixed pipe connection for different types of attachments such as pipes for welding, threaded pipes or grooved pipes. 
Stud bolts	Threaded stud bolts around the port holes secure the flange connections to the apparatus.
Squared loose flange	The Squared loose flange is a special flange supplied by Alfa Laval to be used with the customers piping and is attached with four stud bolts.

Optional components

Foot	Gives stability and is used to secure the PHE with bolts to the foundation. Feet is an optional feature.
Protection sheets	Cover the plate pack and protect against leakage hot or aggressive fluids and the hot plate pack. 
Bolt protection	Plastic tubes that protects the threads of the tightening bolts.
Insulation	For application where the PHE surface will be hot or cold insulation can be used.
Earthing lug	An earthing connection is used to eliminate the risk of static electricity building up in the equipment.
Nozzle cover	Protection to avoid particles to enter into the PHE during transportation.
Drip tray	Depending on the type of fluid in the PHE and the type of installation, a drip tray (drainage box) may be necessary to avoid injury to personnel and damage to equipment.

Name Plate

On the name plate the type of unit, manufacturing number and manufacturing year can be found. Pressure vessel details in accordance with the applicable pressure vessel code are also given. The name plate is fixed to the frame plate, most common, or the pressure plate. The nameplate can be a steel plate or a sticker label.

Warning!
For each unit, the mechanical design pressures and temperatures are marked on the name plate. These must not be exceeded.

Caution!
Avoid aggressive chemicals for cleaning the PHE when sticker label is used.

The mechanical design pressure and the design temperature as given on the name plate are the values to which the plate heat exchanger is approved to the pressure vessel code in question. The mechanical design temperature may exceed the operating temperature for which the gaskets have been selected to reach a its lifetime. If the operating temperatures as specified on the assembly drawing are to be exceeded the supplier should be consulted.

1. Space for logotype.
2. Open space.
3. Web site for service.
4. Drawing of possible locations of connections. For 3A units location of 3A tag.
5. Space for mark of approval.
6. Warning, read manual.
7. Date of pressure test.
8. Maximum operating temperatures.
9. Test pressure.
10. Max permissible operating temperatures.
11. Max permissible operating pressures.
12. Decisive volume or volume for each fluid.
13. Locations of connections for each fluid.
14. Decisive fluid group.
15. Year of manufacture.
16. Serial number.
17. Type.
18. Manufacturer’s name.

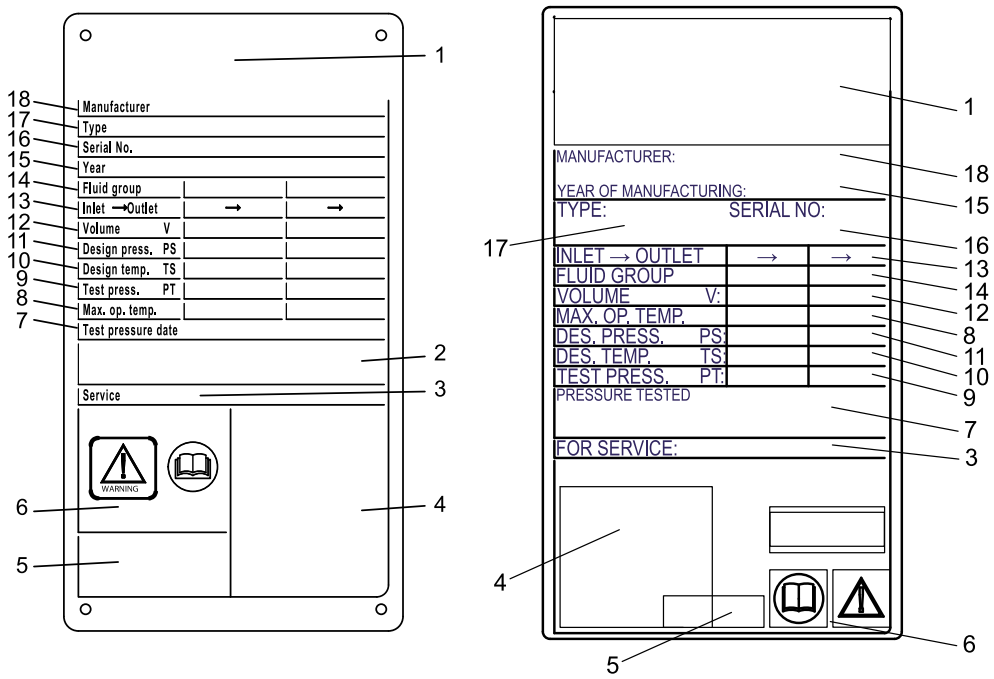


Figure 1. Metallic name plate on the left and label name plate on the right.

Function

The Plate Heat Exchanger (PHE) consists of a pack of corrugated metal plates with port holes for input and output for the two separate fluids. The heat transfer between the two fluids will take place through the plates.

The plate pack is assembled between a frame plate and a pressure plate and compressed by tightening bolts. The plates are fitted with a gasket which seals the channel and directs the fluids into alternate channels. The plate corrugation promotes fluid turbulence and supports the plates against differential pressure.

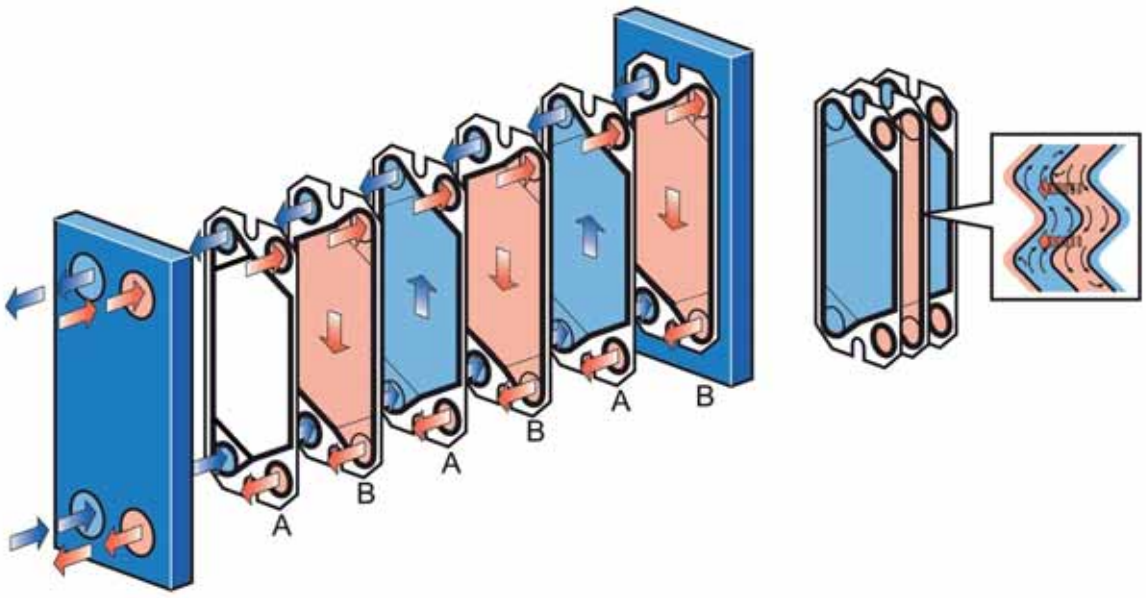


Figure 2. Principle of Plate pack arrangement.

Multi-pass

Multi-pass can be created by using turning plates, with 1, 2 or 3 unholed ports. The main purpose is to change the flow direction of one or both fluids.

For some units partition plate is required to support the unholed ports in the turning plates. Also transition plate need to be added in the package to prevent media to get in contact with the partition or pressure plate.

Example of use for multi-pass can be in processes that requires longer heating periods if the media requires a slower heating.

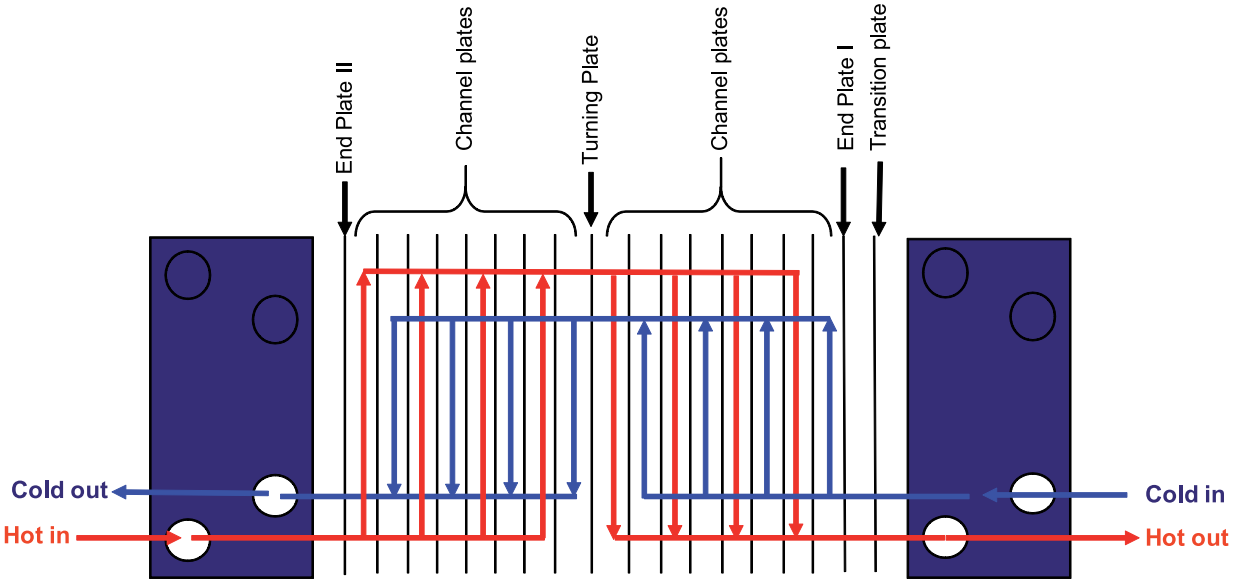
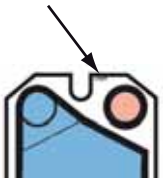


Figure 3. Example of multi-pass set up.

Identification of plate side

The A side of the plate is identified by the stamp with the letter A or the model name, in some cases both, at the top of the plate (refer to figure).



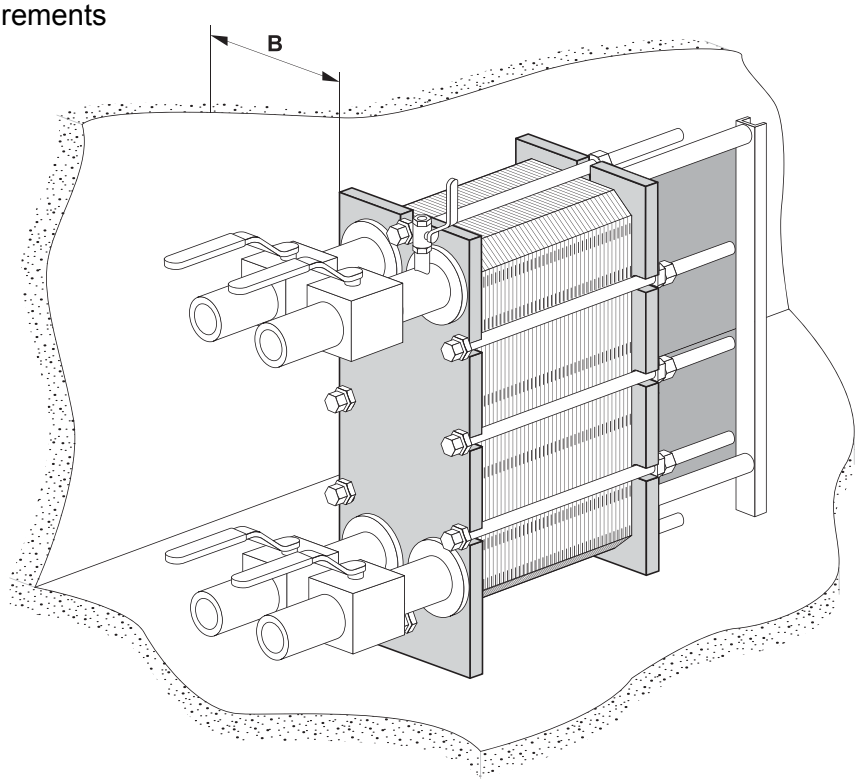
Installation

Before installation

To consider before installation

- Before connecting any piping, make sure all foreign objects have been flushed out of the piping system that should be connected to the PHE.
- Before start-up check that all tightening bolts are firmly tightened and that the correct measurements of the plate pack, refer to PHE drawing.
- When connecting the piping system make sure the pipes do not subject the PHE to stress or strain.
- To avoid water hammer, do not use fast-closing valves.
- Make sure that no air remains within the PHE.
- Safety valves shall be installed according to current pressure vessel regulations.
- It is recommended that protective sheets are used to cover the plate pack. Protect against leakage of hot or aggressive fluids and the hot plate pack
- If the PHE surface temperature is expected to be hot or cold take protective actions, such as insulate the PHE, to avoid risk for personnel injuries. Ensure that required action are according to local regulation.
- For each model, design pressures and temperatures are marked on the identification plate. These shall not be exceeded.

Requirements



Space
A minimum free space is needed for lifting plates in and out. Refer to the delivered PHE drawing.

Foundation
Install on a flat foundation giving enough support to the frame.

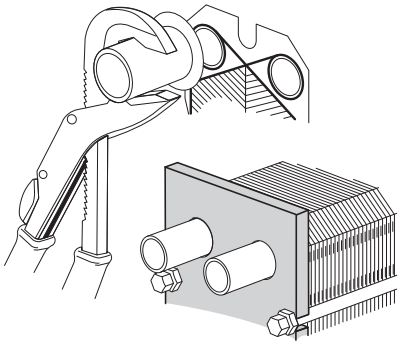
Elbow
To make it easier to disconnect the PHE, an elbow should be fitted to the connection in the pressure plate, directed upwards or sideways, and with another flange located just outside the contour of the Plate Heat Exchanger.

Shut-off valve
To be able to open the PHE, shut-off valves should be provided in all connections.

Connection
Different types of connections can be used to connect the piping system to the apparatus.

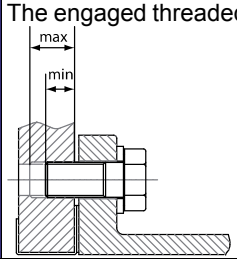
Flanged connections can be attached with either pin bolts or headed bolts.

Make sure the pipe connections are securely held when working on the pipings.



Caution!
Turning of the connections will damage the gaskets on the end plate and cause leakage.

Note!
Using headed bolts require careful measuring to ensure that the bolts get the correct engaged threaded length into the Frame plate and the minimum engagement length (min) are stated in the PHE drawing. The maximum engagement threaded length (max) in the Frame plate is limited, refer to the PHE drawing.
The engaged threaded length must be maintained within the limits for the tightened PHE.



Caution!
Failing to achieve correct engaged threaded length for headed bolt may result in leakage, damage to the frame plate and injury to personnel

Connections in the pressure plate
It is important that the plate pack has been tightened to the correct dimension A (check against PHE drawing) before the piping system is connected.

Drip tray (Optional)
Depending on the type of fluid in the PHE and the type of installation, a drip tray (drainage box) may be necessary to avoid injury to personnel and damage to equipment.

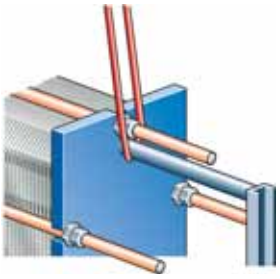
Note!
Put the drip tray in place before positioning the PHE.

Lifting

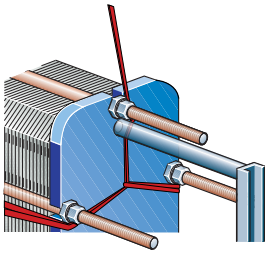
Straps should be used when lifting the PHE. Place straps according to the picture.

Warning!
Never lift by the connections or the studs around them.





Lifting TS-6

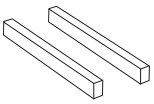


Raising

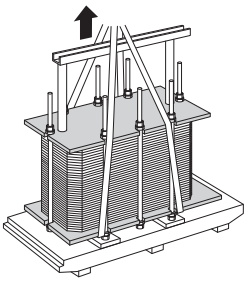
This instruction is valid when raising the PHE after delivery from Alfa Laval. Only use a strap approved for the weight of the PHE. Follow the principle of the instruction below.

Caution!
The straps shall be long enough to be able to rotate the PHE without obstructions. Especially consider the space for the support column.

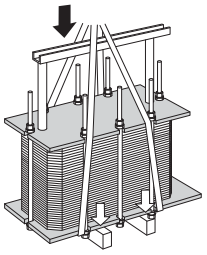
Step 1
Place two timber beams on the floor.



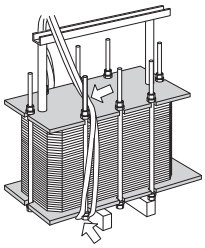
Step 2
Lift the PHE off the pallet using e.g. straps

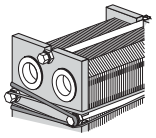


Step 3
Place the PHE on the timber beams.

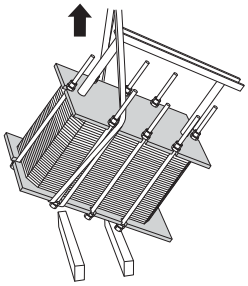


Step 4
Place straps around one bolt on each side.

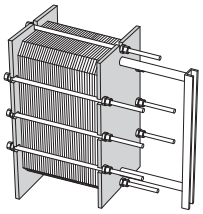




Step 5
Lift the PHE off the timber beams.



Step 6
Lower the PHE into a horizontal position and place it on the floor.



Operation

Start-up

During start-up check that no visible leakages appear from plate pack, valves or piping system.

Caution!
Before pressurizing the PHE it is important to ensure that the temperature of the PHE is within the temperature range stated in the PHE drawing.

Caution!
If the temperature of the PHE is below the minimum temperature for the gaskets prior to the service it is recommended to heat up the PHE above this limit to avoid cold leakage.

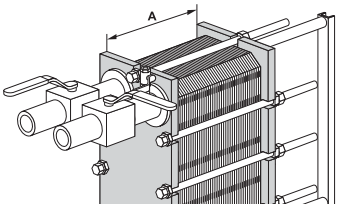
Note!
If several pumps are included in the system, make sure you know which one should be activated first.

Centrifugal pumps must be started against closed valves and valves must be operated as smoothly as possible.

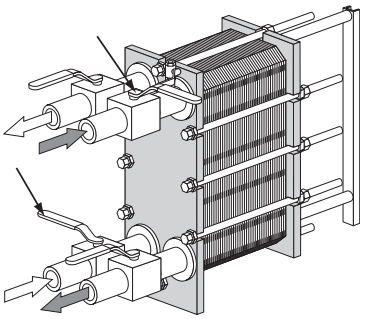
Do not run pumps temporarily empty on the suction side.

Note!
Adjustments of flow rates should be made slowly in order to avoid the risk of pressure surge (water hammer).
Water hammer is a short lasting pressure peak that can appear during start-up or shut-down of a system, causing liquids to travel along a pipe as a wave at the speed of sound. This can cause considerable damage to the equipment.

Step 1
Before start-up check that all tightening bolts are firmly tightened and that the dimension A is correct, refer to PHE drawing



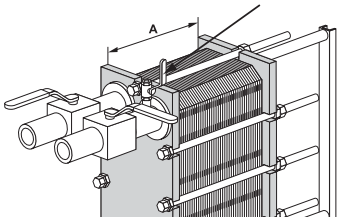
Step 2
Check that the valve is closed between the pump and the unit controlling the system flow rate to avoid pressure surge.



Step 3
If there is a vent valve installed at the exit, make sure it is fully open.

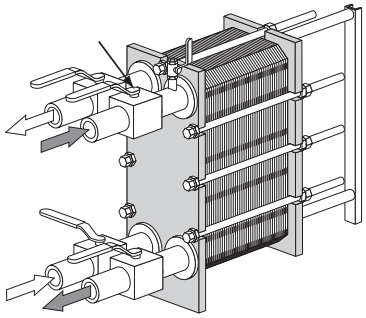
Step 4
Increase the flow rate slowly.

Step 5
Open the air vent and start the pump.

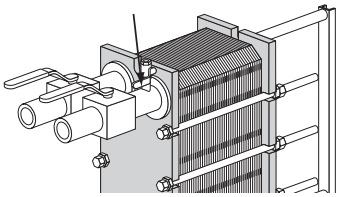


Step 6
Open the valve slowly.

Note!
Avoid rapid temperature changes in the PHE. With media temperatures over 100 °C, slowly increase the temperature preferably at least for one hour.



Step 7
When all air is expelled, close the air vent.



Step 8
Repeat Step 1 – Step 7 for the second media.

Unit in operation

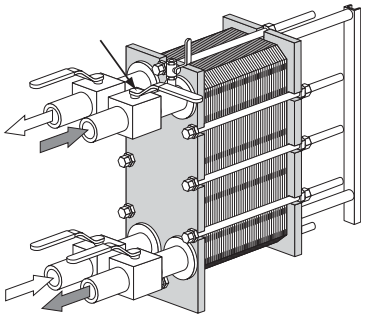
Adjustments of flow rates should be made slowly in order to protect the system against sudden and extreme variations of temperature and pressure.

During operation, check that media temperatures and pressures are within the limits stated on the PHE-drawing and identification plate.

Shut-down

Note!
If several pumps are included in the system, make sure you know which one should be stopped first.

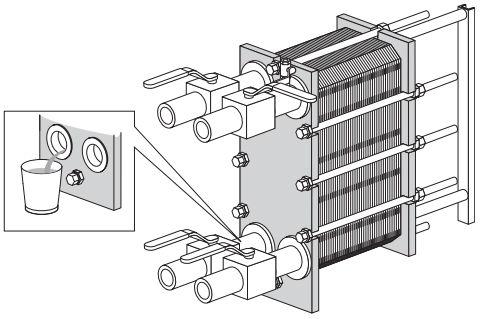
Step 1
Slowly close the valve controlling the flow rate of the pump you are about to stop.



Step 2
When the valve is closed, stop the pump.

Step 3
Repeat Step 1–Step 2 for the other side for the second media.

Step 4
If the PHE is shut down for several days or longer, it should be drained. Draining should also be done if the process is shut down and the ambient temperature is below freezing temperature of the media. Depending on the media processed, it is also recommended to rinse and dry the PHE plates and connections.



Maintenance

To keep the PHE in good condition regular maintenance of the component is required. It is recommended to record maintenance of the PHE.

The plates needs to be cleaned on a regular basis. The frequency depends on several of factors such as type of media and temperatures.

Different methods can be used to perform cleaning. Refer to “Cleaning – Non-product side” on page 18 or a reconditioning at an Alfa Laval service center.

After a longer time of use it can be required to regasketing the PHE by exchanging the gaskets. Refer to “Regasketing” on page 29.

Other maintenance that should be performed regulary:

- Keep carrying bar and guiding bar clean and grease.
- Keep the tightening bolts cleaned and greased.

Cleaning – Non-product side

The Cleaning-In-Place (CIP) equipment permits cleaning of the PHE without opening it. The purpose of cleaning with CIP are as follows:

- Cleaning of fouling and descaling of lime deposits
- Passivation of cleaned surfaces to reduce susceptibility to corrosion
- Neutralization of cleaning liquids before draining.

Follow the instructions of the CIP equipment.

Warning!
Use proper protective equipment, such as safety boots, safety gloves and eye protection, when using the cleaning agents.



Warning!
Corrosive cleaning liquids. Can cause serious injuries to skin and eye!



CIP equipment
Contact Alfa Laval sales representative for size of CIP equipment.



Caution!
The residuals after a cleaning procedure shall be handled according to the local envionmental regulations. After neutralization most cleaning solutions may be drained to the waste water system under the condition that the fouling deposits do not contain heavy metals, other toxic or environmentally dangerous compounds. It is recommended the neutralized chemicals to be analysed for any type of hazardous compounds that were removed from the system prior to disposal.

Cleaning liquids

Liquids	*Description*
AlfaCaus	A strong alkaline liquid, for removing paint, fat, oil and biological deposits.
AlfaPhos	An acid cleaning liquid for removing metallic oxides, rust, lime and other inorganic scale. Contains repassivation inhibitor
AlfaNeutra	A strong alkaline liquid for neutralization of AlfaPhos before drainage.
Alfa P-Neutra	For neutralization of AlfaP-Scale.
Alfa P-Scale	An acidic powder cleaner for removal of primary carbonate scale but also other inorganic scale.
AlfaDescalent	A non-hazardous acidic cleaning agent for removal of inorganic scale.
AlfaDegreaser	A non-hazardous cleaning agent for removal of oil, grease or wax deposits. Additionally prevent foaming when using Alpacon Descaler.
AlfaAdd	AlfaAdd is a neutral cleaning strengthener designed to be used with AlfaPhos, AlfaCaus and Alfa P-Scale. 0.5–1 vol% is added to the total diluted cleaning solution to provide better cleaning results on oily and fatty surfaces and where biological growth occurs. AlfaAdd also reduces any foaming.

If CIP cannot be done, cleaning must be done manually.
Refer to Manual cleaning of opened units on page 24.

Chlorine as a growth inhibitor
Chlorine, commonly used as growth inhibitor in cooling water systems, reduces the corrosion resistance of stainless steels (including high alloys like Alloy 254 SMO).

Chlorine weakens the protection layer of these steels making them more susceptible to corrosion attacks then they otherwise should be. It is a matter of time of exposure and concentration.

In every case where chlorination of non-titanium equipment cannot be avoided, your local representative must be consulted.

Water of more than 300 ppm Cl ions may not be used for preparation of cleaning solutions.



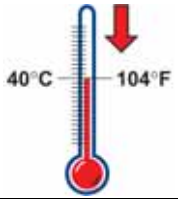
Caution!
Ensure that the handling of residuals after using chlorines follow the local envionmental regulations.

Opening

To perform manual cleaning it is required to open the PHE to clean the plates.

Note!
Before opening the PHE check the warranty conditions. If in any doubt, contact an Alfa Laval sales representative. Refer to “Warranty conditions” on page 4.

Warning!
If the Plate Heat Exchanger is hot, wait until it has cooled down to about 40°C (104°F).

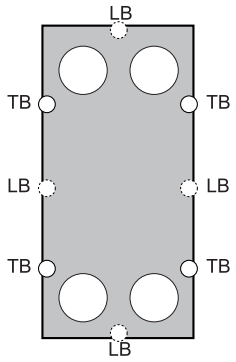


Warning!
If necessary, use proper protective equipment, such as safety boots, safety gloves and eye protection, depending on type of media in the PHE.



Bolt configuration

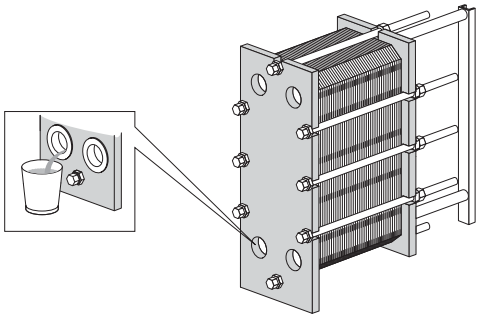
The bolt configuration of the PHE varies between different models. The major force of the plate package is hold by the tightening bolts (TB). To distibute the force evenly over the Frame and Pressure plate locking bolts (LB) are used as well. The locking bolts can be shorter and have smaller dimensions. In the opening and closing procedure it is important to identify the tightening bolts (TB) and the locking bolts (LB). Refer to picture below.



Step 1
Shut down the PHE.

Step 2
Close the valves and isolate the PHE from the rest of the system.

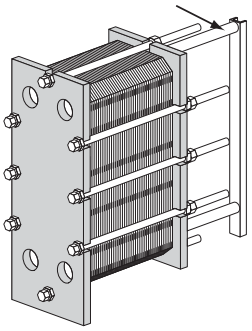
Step 3
Drain the PHE.



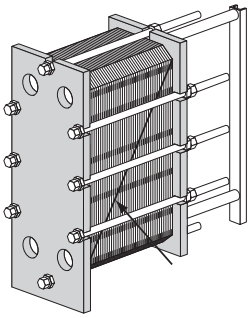
Step 4
Remove the protective sheets, if any.

Step 5
Dismantle pipes from the pressure plate so that the pressure plate are free to move along the carrying bar.

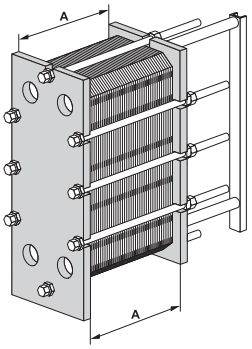
Step 6
Inspect the sliding surfaces of the carrying bar and wipe clean and grease.



Step 7
Mark the plate assembly on the outside with a diagonal line.

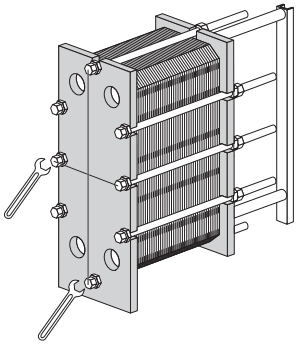


Step 8
Measure and note the dimension A.

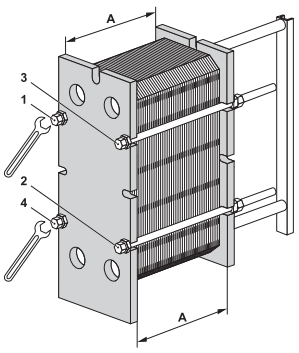


Step 9
Loosen and remove the locking bolts, identify according to on page the bolt configuration prior to this step instruction.

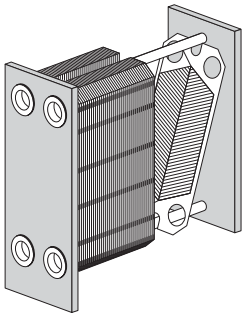
	Note! Brush the threads of the tightening bolts with a steel wire brush and then grease before loosening them.



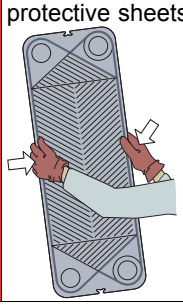
Step 10
Use the tightening bolts to open the PHE. During the opening procedure keep the frame plate and pressure plate always in parallel. Skewing of the pressure plate during opening must not exceed 10 mm (2 turns per bolt) across the width and 25 mm (5 turns per bolt) vertically
Loosen the tightening bolts evenly in the numbered order, 1-2-3-4. Continue alternately until all reaction forces of the plate package has disappeared. Then remove the bolts.



Step 11
Open the plate pack by letting the pressure plate glide on the carrying bar. Usually it is not necessary to remove the plates to clean them.



Caution!
To avoid hand injuries from sharp edges, protective gloves should always be worn when handling plates and protective sheets.



Caution!
When opening the plate pack of the models M3, T2 and TL3, be careful when moving the pressure plate. Make sure the pressure plate is positioned safely from the end of the carrying bar.
A starlock ring (spare part no. 33500045-45) can be attached to the end of the carrying bar to make sure the pressure plate will not pass the end of the carrying bar.

If plates are to be numbered, do this before removing the plates. Plates need not to be removed if cleaning is done using only water, i.e. without cleaning agent.

Warning!
The plate pack may still contain a small residual amount of liquid after draining. Depending on the type of product and type of installation, special arrangements, e.g. drainage box, may be necessary to avoid injury to personnel and damage to equipment.

Manual cleaning of opened units

Caution!
Never use hydrochloric acid with stainless steel plates. Water of more than 330 ppm Cl may not be used for the preparation of cleaning solutions.
It is very important that carrying bars and support columns in aluminium are protected against chemicals.

Note!
Be careful not to damage the gasket during manual cleaning.

Warning!
Use proper protective equipment, such as safety boots, safety gloves and eye protection, when using the cleaning agents.



Warning!
Corrosive cleaning liquids can cause serious injuries to skin and eyes!



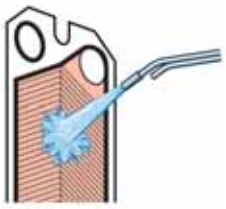
Deposits removable with water and brush
Plates do not need to be removed from the PHE during cleaning.

Step 1
Start cleaning when the heating surface is still wet and the plates are hanging in the frame.

Step 2
Remove deposits using a soft brush and running water.

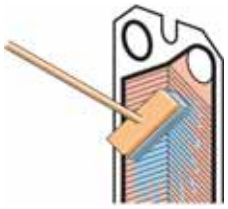


Step 3
Rinse with water using a high pressure hose.



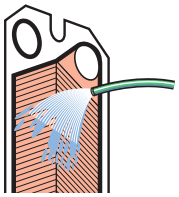
Deposits not removable with water and brush
Plates must be removed from the PHE during cleaning. For a choice of cleaning agents, refer to “Cleaning liquids” on page 19.

Step 1
Brush with cleaning agent.



Step 2
Rinse immediately with water.

> ***Note!***
> Long exposure to the cleaning agents can damage the gasket glue.



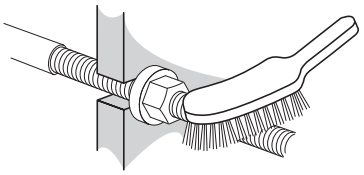
Closing

Follow the instructions below to ensure that the PHE will be properly closed.

For identifying bolt refer to Bolt configuration on page 20

Step 1
Check that all the sealing surfaces are clean.

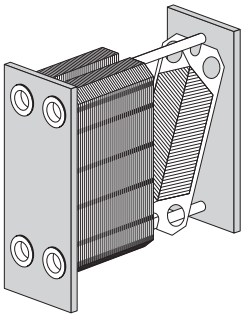
Step 2
Brush the threads of the bolts clean, using a steel wire brush or the Alfa Laval thread cleaner. Lubricate the threads with a thin layer of grease, e.g. Gleitmo 800 or equivalent.



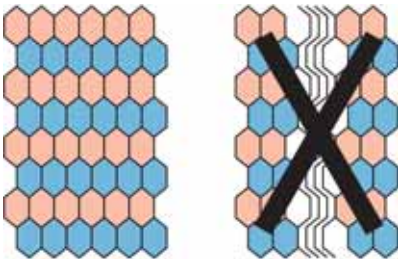
Step 3
Attach gasket to the plates or check that all the gaskets are properly attached.

Note!
If the gasket is wrongly positioned, it will show by the fact that it rises out of the gasket groove or that it is positioned outside the groove.

Step 4
If the plates have been removed insert the them in alternate directions and with the gaskets turned towards the frame plate or pressure plate as specified on the plate hanging list. Use the marked line that was done when the PHE was opened, refer to Step 7 in “Opening” on page 20.

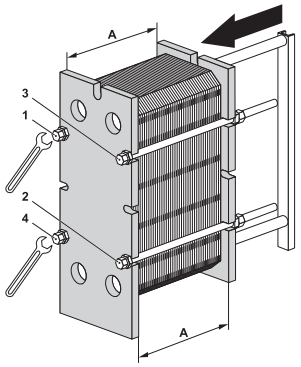


Step 5
If the plate pack has been marked on the outside, check this Step 7 in “” on page 20.
If the plates are correctly assembled (A/B/A/B etc.), the edges form a “honeycomb” pattern, see picture.



Step 6
Press the plate package together. Position the tightening bolts according to the figure.

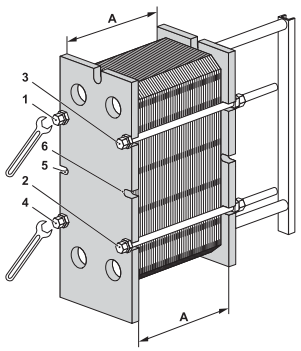
Note!
For TL6-B: Position tightening bolts 1 and 2 (alternatively 3 and 4). Tighten the tightening bolts until the reaction force of the plate package can be noticed. Make sure that the frame and pressure plate are in parallel during the closing procedure (within maximum 10 mm). Position remaining bolts and tighten them to the same dimension.



Tighten the tightening bolts (1-2-3-4) until the reaction force of the plate package can be noticed. Make sure that the frame and pressure plate are in parallel during the closing procedure.

Step 7
Tightening the tightening bolts evenly in the numbered order, 1-2-3-4. Continue alternately until dimension A has been reached.

Note!
For TL6-B: Continue to tighten 10 mm or less with bolt 5 and 6. Then tightening remaining bolts with same length. Repeat those step until the dimension A is achieved.



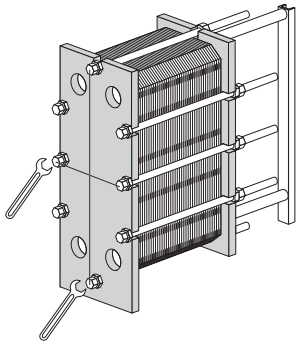
When a pneumatic tightening device is used, see table below for maximum torque. Measure dimension A during tightening.

Bolt size	Bolt with washer	
	Nm	kpm
M10	32	3.2
M16	135	13.5
M20	265	26.5
M24	450	45
M30	900	90

For manual tightening, the tightening torque has to be estimated.
If dimension A cannot be reached
- Check the number of plates and the dimension A.
- Check that all the nuts and bearing boxes are running freely. If not, clean and lubricate, or replace.

Step 8
Position the locking bolts. Tightening the bolts and check measurement A on both sides, at top and bottom

Note!
Not valid for TL6-B.



Step 9
Mount protection sheets (if provided).

Step 10
Connect pipes.

Step 11
If the PHE does not seal when the measure A has been reached, it can be tightened further to A minus 0.5 %.

Note!
For TL6-B: When using frame ASME standard! PHE units with pressure vessel code ASME are equipped with top and bottom bolts. Tighten those bolts after the procedure above has been finished or slightly before the dimension A is reached.

Pressure test after maintenance

Before start-up of production, whenever plates or gaskets have been removed, inserted or exchanged, it is strongly recommended to perform a pressure test to confirm the internal and external sealing function of the PHE. At this test, one media side at a time must be tested with the other side open to the ambient pressure.

Caution!
The pressure testing shall be performed at a pressure equal to the operating pressure of the actual unit but never above the design pressure as stated on the nameplate.

The recommended test time is 10 minutes for each media.

Please consult the local Alfa Laval representative of the supplier for advice on the pressure testing procedure.

Regasketing

The procedures below regards Field gaskets, Ring gaskets and End gaskets

> ***Note!***
> Before removing the old gaskets check how it is attached.

Clip-on

Step 1
Open the PHE, refer to on page and remove the plate that is to have a new gasket.

> ***Note!***
> Before opening the PHE check the warranty conditions. If in any doubt, contact an Alfa Laval sales representative. Refer to “Warranty conditions” on page 4 .

Step 2
Remove the old gasket.

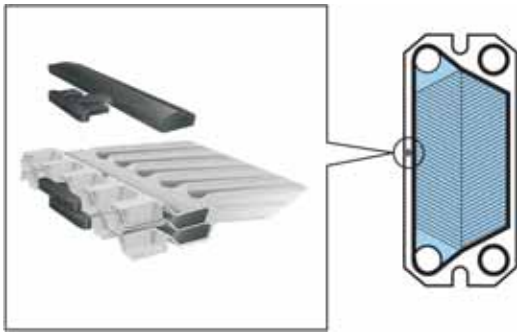
Step 3
Assure that all sealing surfaces are dry, clean and free of foreign matter such as fat, grease or similar.

Step 4
Check the gasket and remove rubber residual before attaching it.

> ***Note!***
> Especially the end plate gasket!!

Step 5
Attach the clip-on gasket to the plate. Slip the gasket tabs under the edge of the plate.

> ***Note!***
> Make sure the two gasket prongs are in correct position.



Step 6
Repeat the procedure until all plates that is needed to be regasketed is done. Close the PHE according to Closing on page 26.

Glued gaskets
Use glue recommended by Alfa Laval. Separate gluing instructions will be delivered together with the glue.

Caution!
Other glues than those recommended can contain chlorides that can damage the plates.

Warning!
Do not use sharp tools when removing the glued gasket to avoid damage to the plates.